

Chapter 45. Oxygen Toxicity

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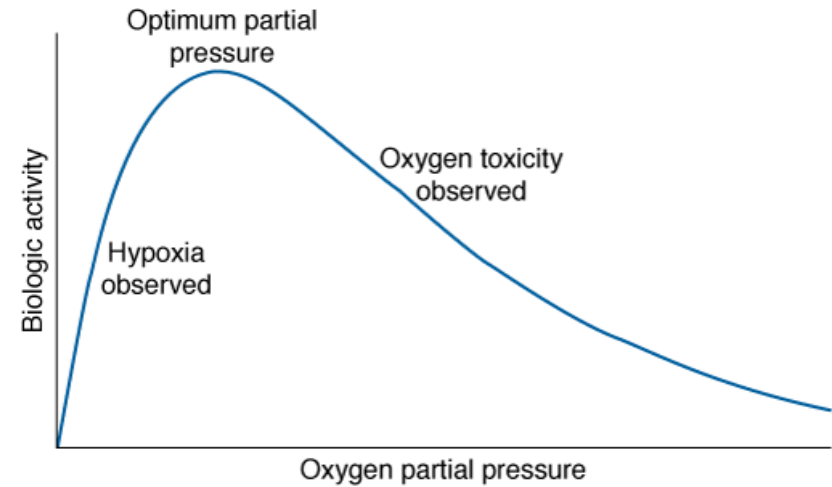
Oxygen Toxicity: Introduction

The importance of the physiology and toxicology of [oxygen](#) (O_2) breathing have increased in recent years. The past 25 years have witnessed a remarkable upsurge of knowledge and interest in “oxidative stress” throughout all of biology. The use of O_2 continues to grow, from critically ill to ambulatory patients and even to recreational use at “[oxygen bars](#).”¹ Recent advances in patient care have refocused attention on the optimum use of O_2 . For example, currently, strategies to protect the lung from mechanical injury during mechanical ventilation emphasize the use of lower tidal volumes. But such strategies may impair gas exchange, resulting in higher requirements for inspired O_2 fraction (FI_{O_2}).²

Lavoisier initially characterized O_2 as “highly respirable air”³ and “vital air” before eventually giving it the name “principle oxygene” (acidifying principle) in 1777. He took the name from the Greek roots “oxy” (acid) and “gen” (to form), because initially O_2 was incorrectly believed to be the essential principle in the formation of acids.⁴ Normobaric hyperoxia may be defined as an inspired O_2 tension, PI_{O_2} , between 160 and 760 torr (i.e., between 0.21 and 1.0 atmosphere [atm] of pressure), whereas hyperbaric hyperoxia denotes that PI_{O_2} is greater than 760 torr.

J.B.S. Haldane⁵ and others⁶ speculated, and it is now generally accepted, that life on earth began anaerobically when the earth’s atmosphere was virtually devoid of O_2 . Gilbert⁶ postulated that in this primordial reducing atmosphere, the first living cells used hydrogen, diffusing into the cell from the environment, as an energy source (e.g., metabolizing carbohydrates to methane and water). Gilbert⁶ speculated further that because hydrogen would also reduce essential cellular constituents and thereby poison the cell, these early cells also had to develop antihydrogen defenses and actively transport hydrogen ions out of the cell. As the atmosphere was transformed from a reducing to an oxidizing one, O_2 replaced hydrogen as an energy source. Therefore, to avoid O_2 poisoning, cells then had to develop antioxygen and antioxidant defenses. These observations emphasize that as an energy source for cells, O_2 has a dual effect: It is both life promoting and life destroying. This dual nature of O_2 ([Fig. 45-1](#)) was noted by Priestley in 1775 shortly after his discovery of O_2 : “though pure [[oxygen](#)] might be very useful as a medicine, ... as a candle burns out much faster in [[oxygen](#)] ... so we might ... *live out too fast*.”⁷ Indeed, even at ambient concentrations, O_2 is now considered to play a role in the natural process of aging.⁸⁻¹⁰ In 1777, Scheele, who independently discovered O_2 , noted along with Priestley that O_2 is toxic to plants. In 1785, Lavoisier, who first recognized the vital role of O_2 in the equivalent processes of respiration and combustion, commented explicitly on the dual nature of O_2 : “when there is an excess of vital air [[oxygen](#)], the animal only undergoes a severe illness; when it is lacking, death is almost instantaneous.”⁶ In experiments carried out with Laplace, he described right-heart and pulmonary congestion in guinea pigs that died in O_2 under a bell jar before the O_2 was used up.^{11,12}

Figure 45-1



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Dual effect of oxygen on biologic activity. (Used, with permission, from Gilbert.⁸)

In 1873, Paul Bert became the first to systematically study O₂ toxicity.¹¹ He documented the toxicity of O₂, particularly the lethality of hyperbaric hyperoxia, in all forms of life from mammals to viruses. He concluded that “no living thing is exempt from damage that can be produced by oxygen.”¹³ Grand mal seizures are the most notorious toxic effect of hyperbaric hyperoxia and have been referred to as the “Paul Bert effect.”¹¹ J.B.S. Haldane experienced such seizures himself.¹¹ The pathophysiologic effects of hyperbaric hyperoxia have been reviewed elsewhere.^{11,13-18} This chapter focuses on the effects of normobaric hyperoxia.

Respiratory Effects of Oxygen Breathing

The best known and most studied effects of normobaric hyperoxia concern the respiratory system. Lavoisier was the first to note changes in the lungs of animals that died from hyperoxia, “...the lung was very flaccid, but bright red, even on the outside, and highly congested with blood....”¹⁹ In 1899, J. Lorrain Smith became the first to systematically study the pulmonary pathology of normobaric hyperoxia, specifically, a 4-day exposure to a F_IO₂ of 0.74 to 0.80 in mammals. The diffuse intense lung damage from O₂ toxicity described by Smith, and many others since, has been named the “Lorrain Smith effect.”¹¹

The symptoms and physiologic changes in healthy human volunteers associated with normobaric hyperoxia are the subject of several reviews.^{11,13,16,17,20-30} Table 45-1 lists the respiratory effects of O₂ breathing.

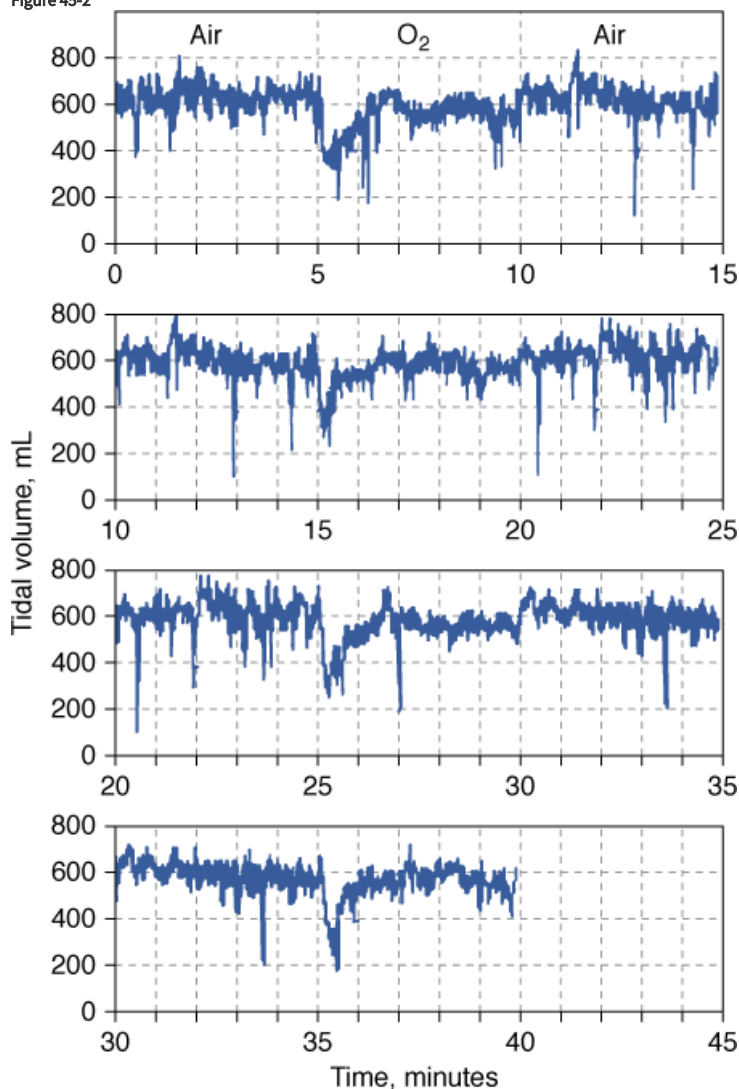
Table 45-1: Respiratory Effects of Oxygen Breathing (Normobaric Hyperoxia)

Depression of respiration
Stimulation of respiration
Pulmonary vasodilation; ventilation–perfusion mismatching
Hypercapnia
Absorption atelectasis
Acute tracheobronchitis; decreased mucociliary clearance
Diffuse alveolar damage; adult respiratory distress syndrome
Bronchopulmonary dysplasia

Respiratory Depression and Stimulation, Pulmonary Vasodilation, and Hypercapnia

The depression in respiratory drive is primarily related to decreased stimulation of the hypoxia-sensitive chemoreceptors in the carotid and aortic bodies. These chemoreceptors are the only known O_2 sensors that initiate chemoreflexes.^{31,32} Dejours et al showed that the O_2 chemoreceptor drive of respiration, which may account for 10% to 15% of resting ventilation, disappears above an alveolar partial pressure of oxygen (P_{O_2}) of 170 torr, corresponding to a P_{iO_2} of 230 torr, or an F_{iO_2} of 0.30.¹⁶ When O_2 breathing is suddenly initiated, the normal response is twofold.^{21,33,34} First, there is a nearly immediate, precipitous decrease in ventilation, whose nadir occurs within 20 to 30 seconds (Fig. 45-2), followed by a gradual return to and slightly above baseline (Fig. 45-3). The acute transient depression is mediated by the arterial chemoreceptor reflexes and may be taken as an index of the level of chemoreceptor activity present during normoxia.^{33,35} The subacute mild sustained ventilatory stimulation appears to be secondary to an indirect effect of hyperoxia on the respiratory centers in the brainstem. That is, hyperoxia causes an increase in tissue carbon dioxide tension (P_{CO_2}) secondary to the Haldane effect, or Christian-Douglas-Haldane effect³⁶ (the increase in blood P_{CO_2} , at a fixed CO_2 content, caused by the release of hemoglobin-bound hydrogen ions and CO_2 , bound as carbamino compounds, that occurs on the oxygenation of hemoglobin). Ironically, in one of the earliest reports (1921) of this hyperoxia-induced mild hyperventilation, Dautrebande and Haldane³⁷ proposed that hyperoxic vasoconstriction of the central nervous system vasculature produces tissue hypercapnia, which, in turn, directly stimulates the respiratory center.

Figure 45-2

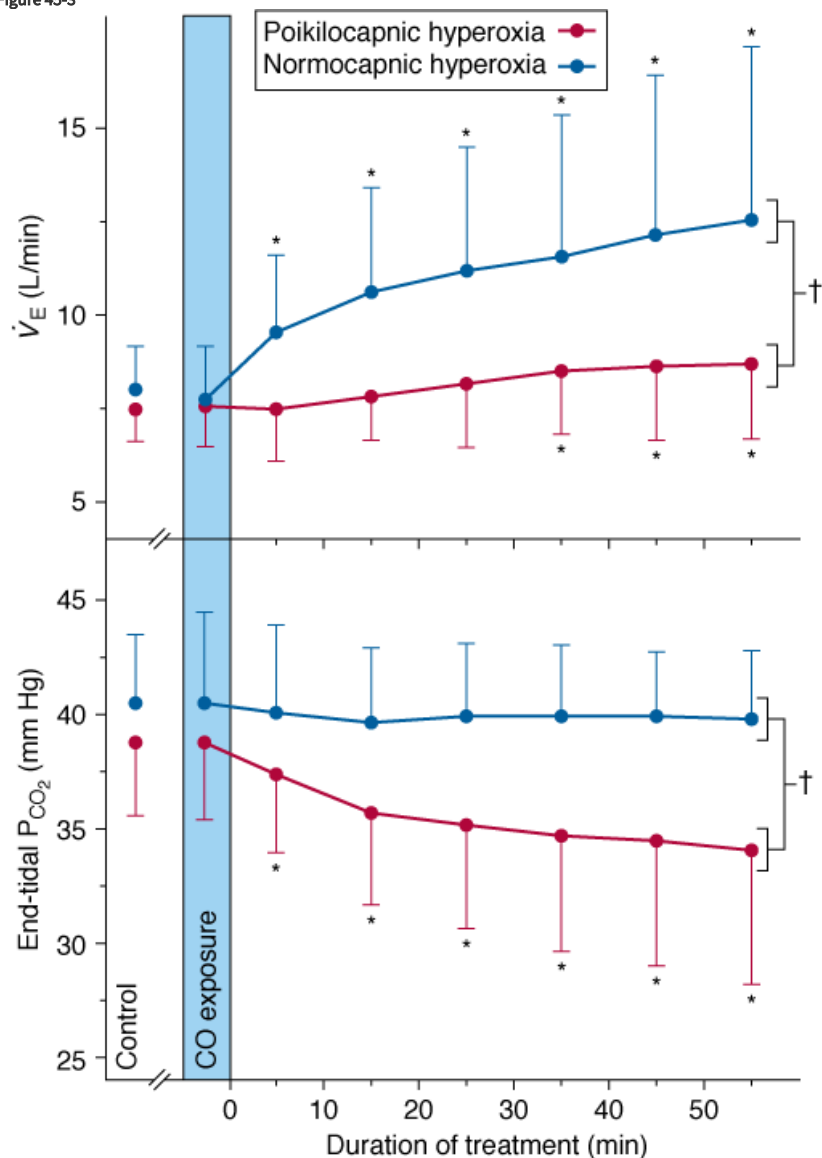


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Acute depression of respiration by normobaric hyperoxia illustrated in a healthy conscious dog breathing from a mask as the inspired gas was alternated between air and O_2 every 5 minutes for 40 minutes. At the onset of each period of O_2 breathing, the continuous

record of tidal volume shows a precipitous fall to a mean value of 54% of the normoxia baseline. This decrease in tidal volume is evident by 10 seconds and reaches a nadir at approximately 18 seconds. It is transient, and despite continued hyperoxia, it has returned nearly to baseline by 1.5 minutes. The record is plotted such that the normoxia data at the end of each 15-minute period are repeated on the next line as the beginning of the next 15-minute period to emphasize the repeating, cyclic nature of the alterations in inspired gas and to demonstrate more clearly the response pattern of tidal volume to hyperoxia. (Used, with permission, from Lodato and Jubran.³³)

Figure 45-3



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Steady-state responses of minute ventilation ($\dot{V}_E$) and end-tidal partial pressure of carbon dioxide (P_{CO_2}) to breathing 100% O_2 (hyperoxia) in healthy human subjects (mean $\pm$ standard deviation [SD]; $n = 14$). Poikilocapnic hyperoxia (end-tidal P_{CO_2} was not controlled) produced a mild increase in minute ventilation and a small (4 mm Hg) decrease in end-tidal P_{CO_2} . Normocapnic hyperoxia (CO_2 was added to the inspire to maintain constant end-tidal P_{CO_2}) produced a marked and progressive increase in minute ventilation. These respiratory stimulant effects of hyperoxia are due largely to the Haldane effect, which results in increased P_{CO_2} in the brain. *Note:* Before time = 0, subjects were exposed to carbon monoxide in this study of CO elimination. (Used, with permission, from Rucker et al.⁴¹)

In normal subjects the mild respiratory stimulation induced in the steady state by hyperoxia results in a mild hyperventilation and a mild decrease in partial pressure of arterial carbon dioxide (Pa_{CO_2}) by a few millimeters of mercury, which, in turn, limits the degree of hyperventilation. But in the presence of inspired CO_2 , the paradoxical respiratory stimulant effect of O_2 is much more obvious, as noted

as early as 1918 by Yamada.³⁸ When the normally modest fall in P_{aCO_2} is prevented, (isocapnic) hyperoxia produces a dramatic increase in ventilation (see Fig. 45-3),^{39–41} by 21% at $F_{IO_2} = 0.30$ and by a remarkable 115% at $F_{IO_2} = 0.75$.^{39,40} Becker et al.⁴⁰ showed that in both poikilocapnic (nonisocapnic) and isocapnic hyperoxia, the Haldane effect is the most important mechanism of the respiratory stimulant effect of hyperoxia in normals. Rucker et al.⁴¹ used these observations and showed in normal subjects that, compared with poikilocapnic hyperoxia, isocapnic hyperoxia more rapidly eliminates carbon monoxide and improves cerebral blood flow and O_2 delivery (secondary to the vasodilator effect of the higher level of P_{aCO_2}) (see Fig. 45-3).

In contrast to the steady-state respiratory stimulant effect of hyperoxia in normals, hyperoxia may instead cause severe respiratory depression in patients with severe chronic obstructive pulmonary disease (COPD). The relative importance of the various mechanisms remains controversial. The conventional view is that hyperoxic hypercapnia is caused by alveolar hypoventilation resulting from reduced hypoxic ventilatory drive secondary to the rise in arterial P_{O_2} (P_{aO_2}),⁴² analogous to “ O_2 apnea” in patients recovering from general anesthesia.¹⁷ Other mechanisms include the Haldane effect, deterioration of ventilation–perfusion ($\dot{V}/\dot{Q}$) matching in the lung secondary to hyperoxic release of hypoxic pulmonary vasoconstriction, and development of atelectasis (true shunt). O_2 may also impair $\dot{V}/\dot{Q}$ matching by release of hypoxia-induced bronchoconstriction.^{43,44}

The data in both acutely decompensated⁴⁵ and stable^{46–48} patients with COPD, however, indicate that deterioration of $\dot{V}/\dot{Q}$ matching is the predominant mechanism. Using the multiple inert gas elimination technique in COPD patients, Wagner et al.⁴⁹ found no systematic changes in the distribution of $\dot{V}/\dot{Q}$ ratios with 100% O_2 . But they did not report attempts to correlate individual changes in P_{aCO_2} and changes in $\dot{V}/\dot{Q}$ distribution. In mechanically ventilated patients, Dunn et al.⁵⁰ reported that hyperoxia both worsened $\dot{V}/\dot{Q}$ matching and depressed respiratory drive. For an informative interchange, see Stradling⁵¹ and Aubier et al.⁵²

In a more recent study of patients with exacerbations of COPD, Robinson et al.⁴⁴ used the multiple inert gas elimination technique to compare those patients who retained CO_2 with O_2 with those who did not. In both groups, hyperoxia worsened $\dot{V}/\dot{Q}$ matching by increasing perfusion to relatively poorly ventilated lung units (release of hypoxic vasoconstriction). In only the group who retained CO_2 , however, hyperoxia decreased minute ventilation (by 20%) and increased ventilation to overventilated units (increased alveolar dead space). Later, they⁵³ concluded that in the group that retained CO_2 , the largest contribution (46%) was from a decrease in minute ventilation, followed closely (43%) by an increase in alveolar dead space, with a much smaller contribution (11%) from the Haldane effect. The modest Haldane effect was likely secondary to only the modest degree of baseline hypoxemia (mean $P_{aO_2} = 55$ mm Hg). They also showed that the risk of hyperoxic hypercapnia is better predicted by a low baseline P_{aO_2} than by the baseline P_{aCO_2} .⁴⁴

Neff and Petty⁵⁴ showed that severe *chronic* respiratory acidosis (mean of P_{CO_2} 90 torr, mean pH of 7.32) in patients with COPD is well tolerated. Acute hyperoxic respiratory depression can be avoided by using controlled low-flow O_2 administration.^{55,56} Hyperoxic hypercapnia may also occur in neuromuscular disease⁵⁷ and near-fatal asthma.^{58,59}

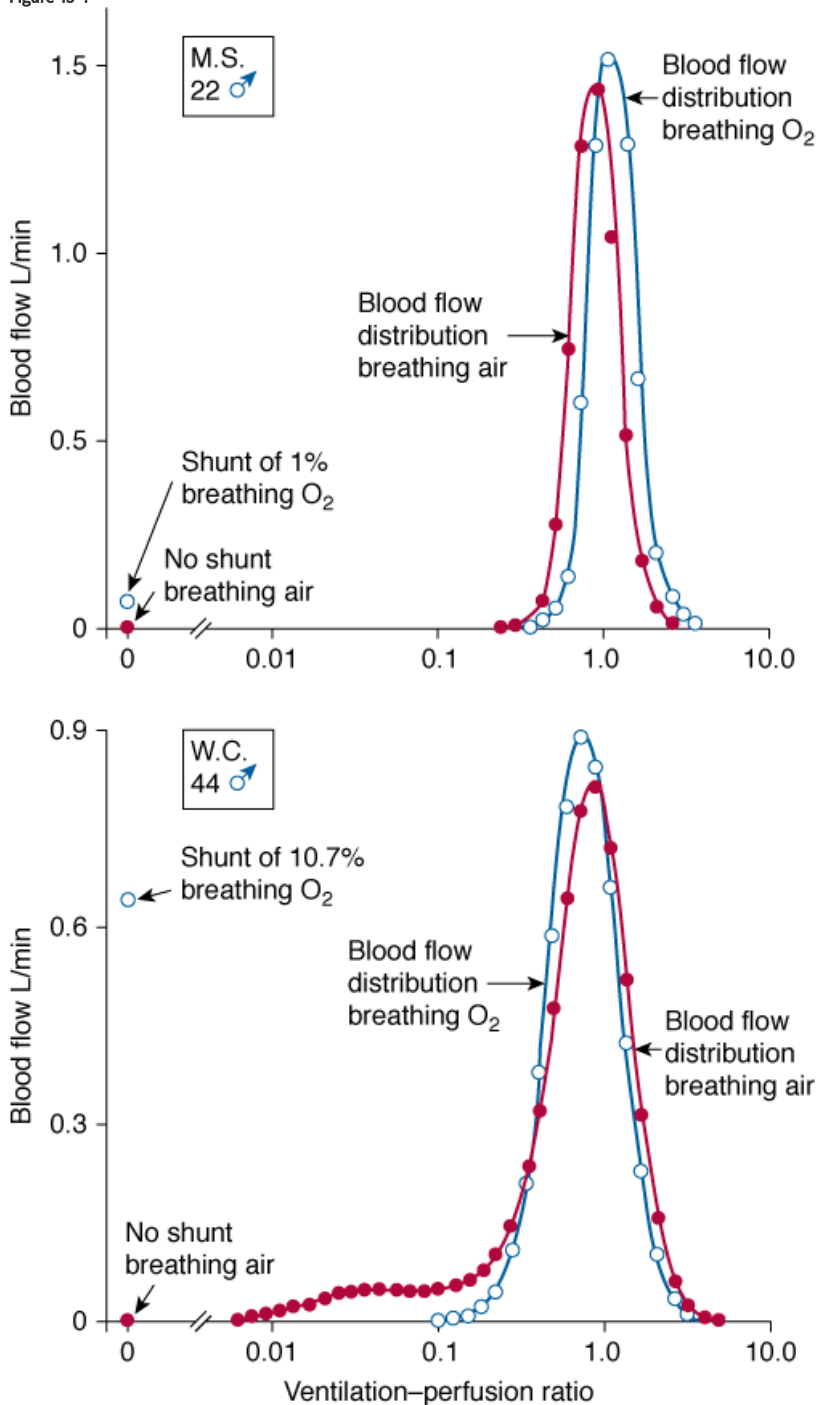
In a prospective randomized controlled trial⁶⁰ of resuscitation of asphyxiated newborn human infants, those resuscitated with room air had better outcomes (Apgar scores, time to first breath, and first cry) than those resuscitated with 100% O_2 . These results were attributed, in part, to the respiratory depressant effect of O_2 .⁶⁰ The current recommendation is that room air is preferred over 100% O_2 for newborn resuscitation.⁶¹

Absorption Atelectasis

Absorption atelectasis during O_2 breathing occurs in lung units with sufficiently low $\dot{V}/\dot{Q}$ ratios.⁶² In such units the rate of capillary absorption of O_2 exceeds the rate of replenishment of alveolar gas during inspiration. This effect depends on the $\dot{V}/\dot{Q}$ ratio, the pattern

of ventilation (e.g., the presence of sighs), the, the duration of the O_2 exposure, the intrinsic stability of the lung units (e.g., tissue and surfactant factors), and the sensitivity of the local hypoxic pulmonary vasoconstriction to hyperoxic release (which acts synergistically by further lowering the $\dot{V}_A/\dot{Q}$ ratio). Using the inert-gas elimination technique in healthy nonsmoking subjects, Wagner et al⁶² showed that within 30 minutes of breathing 100% O_2 , eight of the nine subjects had developed small shunts, averaging 0.5% to 3.2%, with the largest at 10.7% (Fig. 45-4). Theoretically, only approximately 6 minutes is required for the development of such shunts.⁶³ Clearly, when 100% O_2 breathing is used to quantify a shunt, its magnitude might be overestimated compared to that present during air breathing.

Figure 45-4



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Distributions of blood flow with respect to ventilation-perfusion ($\dot{V}_A/\dot{Q}$) ratios in two normal subjects while breathing air (normoxia) and after 30 minutes of breathing 100% O_2 (normobaric hyperoxia). *Upper panel.* In the younger subject hyperoxia produced (a) a 1% shunt, indicating a minor degree of hyperoxia-induced atelectasis, and (b) an overall shift of the main body of the distribution to the right to

higher $\dot{V}_A/\dot{Q}$ ratios, indicating mild hyperoxia-induced hyperventilation in the steady state. *Lower panel.* In the older subject hyperoxia produced more dramatic changes: The left “shoulder” of the normoxic blood flow distribution (i.e., the blood flow distributed to lung units with low $\dot{V}_A/\dot{Q}$ ratios) was converted by hyperoxia to a 10.7% shunt, indicating substantial amounts of hyperoxia-induced atelectasis, but the main body of the distribution was little altered. (Used, with permission, from Wagner et al.⁶²)

Hyperoxia also enhances absorption from other gas spaces: middle ear and sinuses (causing headache), and therapeutically in pneumothorax, bowel obstruction or ileus, and subcutaneous emphysema.¹⁷

Hyperoxia may also cause atelectasis by interfering with the pulmonary surfactant system,⁶⁴ both by injury to the alveolar type II pneumocyte, which synthesizes, secretes, and recycles surfactant, and by injury to the alveolar–capillary interface, resulting in the influx of plasma proteins, which inhibit surfactant function. Hyperoxia-induced atelectasis has also been demonstrated in patients with acute lung injury⁶⁵ using the multiple inert-gas elimination technique.

Acute Tracheobronchitis

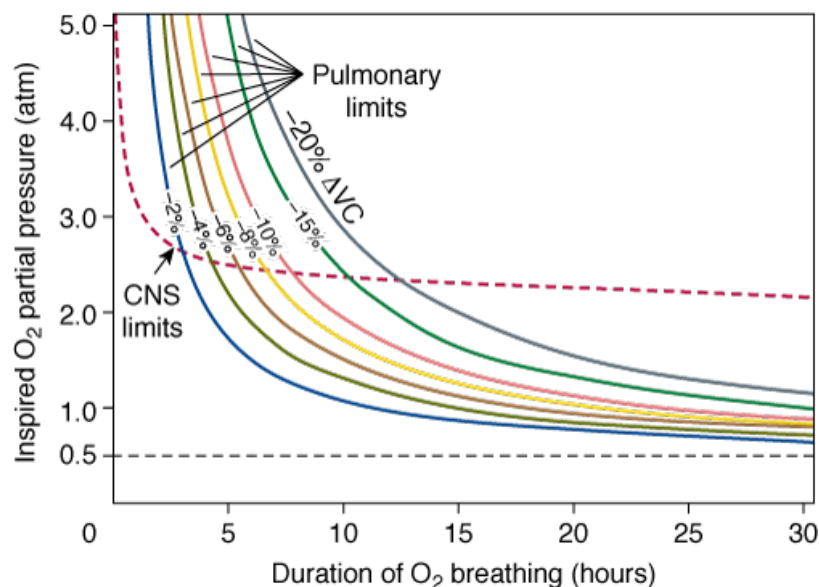
Comroe et al⁶⁶ described acute tracheobronchitis in 1945. Normal subjects breathing 100% O₂ for 24 hours noted substernal distress, cough, sore throat, nasal congestion, eye irritation, ear discomfort, fatigue, and paresthesia, and had a reduction in vital capacity. Symptoms began within 4 and 22 hours after initiating oxygen breathing. Substernal distress was also noted while breathing 75% O₂, but not 50% O₂, for 24 hours. Substernal distress is generally the first symptom noted in normal subjects. It is thought to represent acute tracheobronchitis,²⁶ but it may also be the result of atelectasis alone.⁶⁷ Using fiber-optic bronchoscopy, Sackner et al⁶⁸ directly observed evidence of such tracheobronchial inflammation (focal areas of redness, edema, and injection of small vessels of the trachea, depression of mucus velocity) in healthy human subjects after 6 hours of breathing 90% to 95% O₂.

A decrease in vital capacity⁶⁶ is considered the best index of O₂ toxicity. In humans, this decrease in vital capacity probably results from the inspiratory pain of the acute tracheobronchitis and from absorption atelectasis. In animals, it is more clearly related to parenchymal lung injury. By impairing an important host defense mechanism, acute tracheobronchitis could predispose to nosocomial pneumonia,^{68,69} but this has not been well studied. In rats, hyperoxia also impairs alveolar macrophage function,⁷⁰ perhaps adding to the risk of ventilator-associated pneumonia. In healthy subjects with moderate responsiveness to methacholine at baseline, breathing 95% O₂ for 12 hours produced no change in methacholine responsiveness, despite signs and symptoms of acute tracheobronchitis.⁷¹

Threshold Oxygen Tension for the Development of Oxygen Toxicity

Like other toxic drugs, the effects of O₂ can be viewed in terms of a classical pharmacologic dose–response curve.¹³ The dose of O₂ can be expressed as the product of and the duration of the exposure (i.e., dose = $P_{iO_2} \times \text{time}$). When plotted, this relationship is a rectangular hyperbola (Fig. 45-5), which expresses the dose–response in terms of vital capacity decrements. The horizontal asymptote of 50% O₂, or 0.5 atm, assumes that at that level of hyperoxia, individuals can be exposed safely for prolonged periods of time.

Figure 45-5



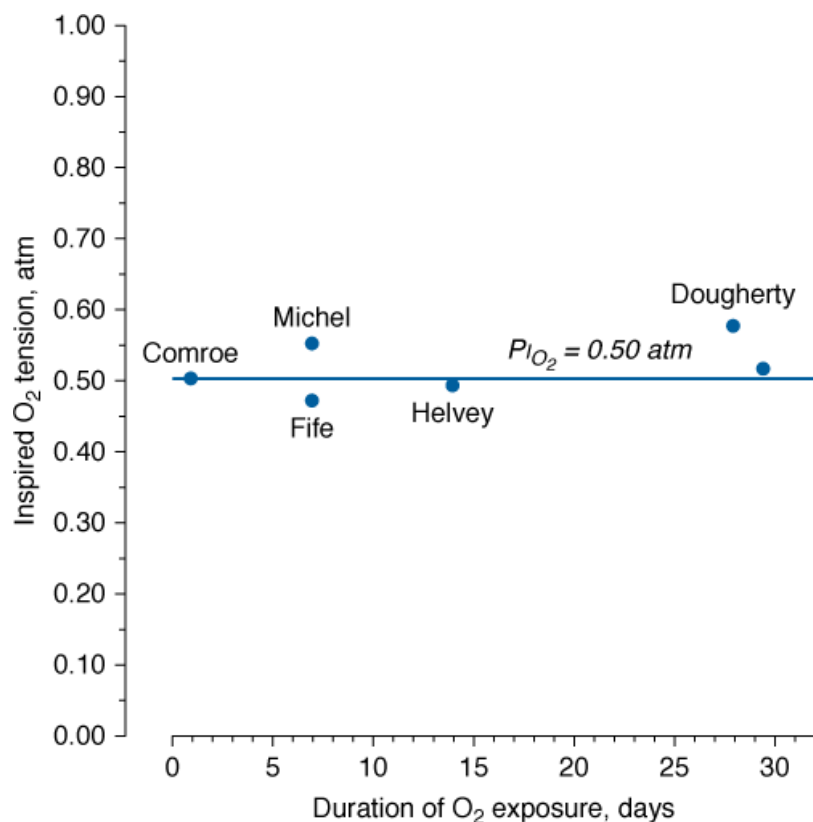
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Oxygen tolerance or dose-response curves for pulmonary and central nervous system (CNS) toxicity in healthy human subjects. Each curve in the family of curves for pulmonary O₂ toxicity is a rectangular hyperbola and represents equivalent O₂ partial pressure-duration “doses” of O₂ exposures. These curves are based on the development of the indicated changes in vital capacity in 50% of the individuals subjected to the indicated O₂ partial pressure-duration exposures. The horizontal asymptote for the lowest pulmonary toxicity curve is at an O₂ partial pressure of about 0.6 atmospheres (atm), indicating that O₂ partial pressures below 0.6 atm appear to be “safe” for exposures of indefinite duration. (Used, with permission, from Lambertsen.³⁰⁶)

Carpagnano et al.⁷² reported that breathing 28% O₂ for only 1 hour increased inflammatory markers (8-isoprostane and interleukin-6) in exhaled breath condensate in both healthy subjects and patients with COPD. Vital capacity, however, did not change in either group. These results were likely a manifestation of subclinical tracheobronchitis (see above).

The validity of setting at 0.5 atm the threshold P_{iO_2} for the development of O₂ toxicity, as reflected by a decrease in vital capacity, for exposures of indefinite duration was carefully discussed by Clark and Lambertsen²¹ and Clark.²² Earlier these authors had suggested a less-conservative threshold of 0.6 atm.⁷³ The rationale for the 0.5-atm P_{iO_2} threshold is based largely on the responses of vital capacity to prolonged hyperoxia in healthy men in the three studies of Comroe et al.,⁶⁶ Helvey et al.,⁷⁴ and Michel et al.⁷⁵ Figure 45-6 summarizes these exposures and those of two subsequent studies in healthy men.^{76,77} Comroe et al.⁶⁶ exposed nine normal men to a P_{iO_2} of 0.75 atm for 24 hours, with return to air breathing for 15 minutes every 3 hours. They found that five of the nine had chest symptoms and that the group had a modest decrease in vital capacity (mean decrease < 300 mL). None of six subjects exposed to P_{iO_2} = 0.5 atm for 24 hours had chest symptoms, and the mean decrease in vital capacity was even less (only approximately 210 mL). They concluded that the safe limit of P_{iO_2} lies between 0.5 and 0.75 atm, probably close to 0.6 atm.⁶⁶ Similar findings were reported by Ohlsson⁷⁸ (not plotted in Fig. 45-6) in six men exposed to a P_{iO_2} of 0.78 to 0.88 atm for 53 to 57 hours. Helvey et al.⁷⁴ found no change in vital capacity, P_{aO_2} , or chest radiographs after 14 days of exposure to 0.49 atm. Michel et al.⁷⁵ reported no net change in vital capacity during a 7-day exposure of six normal U.S. Navy men to P_{iO_2} = 0.55 atm. Fife et al.⁷⁶ studied three subjects at P_{iO_2} of 0.47 atm for 7 days; none developed chest symptoms or a change in vital capacity.

Figure 45-6



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A summary of studies of prolonged exposure of normal human subjects to an inspired O₂ tension ($P_{I_{O_2}}$) near 0.5 atm. The points indicate the $P_{I_{O_2}}$ and duration of exposure for the respective studies. The name associated with each point indicates the original reference. Each of these studies reported that the hyperoxic exposure was generally well tolerated, with little or no symptoms of chest tightness, little or no decrease in vital capacity or other pulmonary function tests, and no long-term sequelae. See text for details. (Data from Comroe et al,⁶⁶ Helvey et al,⁷⁴ Michel et al,⁷⁵ Fife et al,⁷⁶ and Dougherty et al.⁷⁷)

At $P_{I_{O_2}} = 0.5$ atm for 45 hours, Griffith et al⁷⁹ found increased pulmonary clearance of an inhaled tracer and increased albumin in bronchoalveolar-lavage fluid, suggesting the possibility of injury to the alveolar-capillary barrier, or alternatively, only inflamed airways.⁸⁰

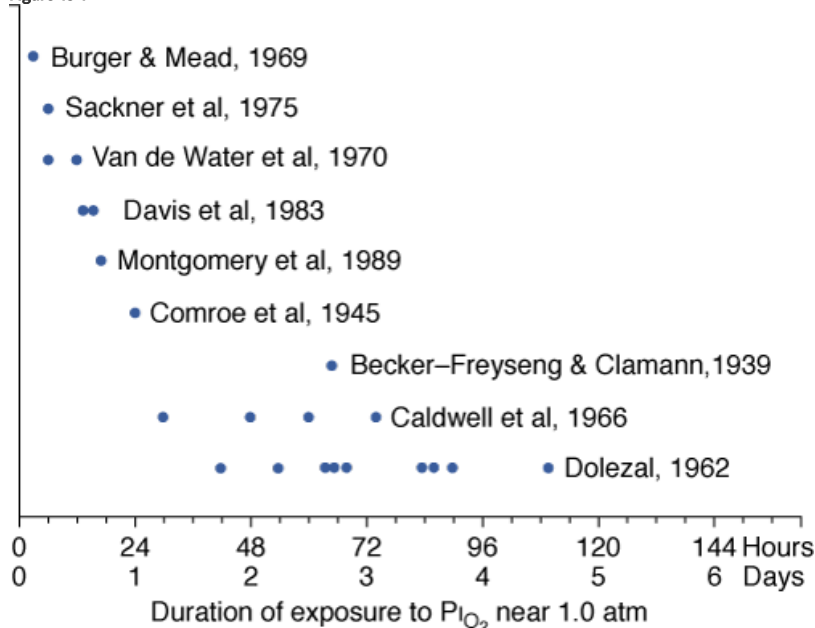
Data from the U.S. Navy's Shallow Habitat Air Diving (SHAD) program support 0.6 atm $P_{I_{O_2}}$ as safe for prolonged O₂ exposure.¹⁶ Men tolerated well exposures of 0.51 and 0.57 atm $P_{I_{O_2}}$ for 29.5 and 28 days, without changes in pulmonary function.^{16,77,81} With 9-day exposures to mean $P_{I_{O_2}}$ of 0.61 atm (each day, 16 hours at $P_{I_{O_2}} = 0.51$ atm and 8 hours at $P_{I_{O_2}} = 0.81$ atm), only one of the three men showed decreased vital capacity (9.7%) and chest discomfort, both coinciding with the daily 8-hour excursions to $P_{I_{O_2}} = 0.81$ atm.^{16,77,81}

Toxicity of Oxygen Near 1 Atmosphere in Normal Subjects

Several studies examining physiologic parameters and other indices of lung injury have also been carried out in humans with exposures at or near a $P_{I_{O_2}} = 1$ atm (Fig. 45-7). Burger and Mead⁶⁷ reported chest pain and tightness and decreases in pulmonary compliance after only 3 hours of breathing pure O₂ at various ambient pressures including 1 atm. The subjects were instructed to refrain from taking deep breaths, yawning, or sighing during the exposure. Postexposure the decrease in compliance and the chest symptoms readily resolved on deep breathing or sighing, suggesting to them that atelectasis rather than direct O₂ toxicity was the etiology. Remarkably, in this study,⁶⁷ breathing pure O₂ at only 0.39 atm ambient pressure ($P_{I_{O_2}}$ of only 0.39 atm) produced symptoms that were qualitatively and quantitatively no different from breathing pure O₂ at 1 atm ambient pressure ($P_{I_{O_2}} = 1$). Yet, breathing with a $P_{I_{O_2}}$ of 1 atm, but in the

presence of an equal amount of nitrogen ($P_{I_{N_2}} = 1$ atm, and total ambient pressure = $P_{I_{O_2}} + P_{I_{N_2}} = 2$ atm) was virtually no different from control (air breathing at 1 atm ambient pressure). These findings further convinced these investigators that atelectasis, promoted by the lack of nitrogen, was the mechanism of both the chest symptoms and the compliance changes.

Figure 45-7



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A summary of studies of prolonged exposure of normal human subjects to an inspired O_2 tension ($P_{I_{O_2}}$) near 1 atm. The horizontal arrays of points indicate the various durations of exposure within a given study. The names associated with each horizontal array of points indicate the original references. See text for details. (Data from Burger and Mead,⁶⁷ Sackner et al,⁶⁸ Van de Water et al,⁸² Davis et al,⁸³ Montgomery et al,⁸⁰ Comroe et al,⁶⁶ Becker-Freyseng and Clamann,⁸⁴ Caldwell et al,⁸⁵ and Dolezal.⁸⁸)

As already noted, Sackner et al⁶⁸ found bronchoscopic evidence of tracheitis in each of ten normal subjects inspiring a $P_{I_{O_2}}$ of 0.90 to 0.95 atm for 6 hours. None had substernal distress or changes in pulmonary function tests. Tracheal mucus velocity was depressed at 3 hours but was restored by a β_2 -adrenergic agonist. They speculated that the effect on mucus velocity could predispose to bacterial superinfection.⁶⁸ In men exposed to 6–12 hours of $P_{I_{O_2}} = 1$ atm and taking a deep breath every 20 minutes, Van de Water et al⁸² found no chest symptoms or changes in pulmonary function or oxygenation.

In fourteen subjects exposed to a $P_{I_{O_2}}$ of 0.95 atm for a mean of 17 hours, Davis et al⁸³ found that nine had substernal discomfort and six had “mild erythema” on bronchoscopy. Bronchoalveolar lavage fluid showed increased albumin, transferrin, and total protein, consistent with alveolar–capillary leak, but no change in inflammatory cells. In contrast, despite chest pain in subjects breathing $P_{I_{O_2}} = 1$ atm for 17 hours, Montgomery et al⁸⁰ found no changes in pulmonary function tests, solute permeability, or systemic or pulmonary endothelial injury (as reflected by plasma fibronectin and factor VIII). They suggested that the increased albumin and transferrin in bronchoalveolar lavage fluid reported by Davis et al⁸³ and Griffith et al⁷⁹ may not indicate lung parenchymal injury, but may instead have come from the inflamed airways.

Comroe et al⁶⁶ studied men exposed to $P_{I_{O_2}} = 0.99$ atm for 24 hours and found no changes in chest radiographs or oxygenation. Burger and Mead⁶⁷ found that breathing pure O_2 at ambient pressure of 0.5 atm, such that $P_{I_{O_2}} = 0.5$ atm and $P_{I_{N_2}} = 0$, produced no symptoms but decreased vital capacity.

In 1939 Becker-Freyseng and Clamann⁸⁴ reported exposures of two men to a $P_{I_{O_2}}$ of 0.9 atm for 65 hours, or 2.7 days. Both had decreased vital capacity. One developed nausea, vomiting, tachycardia, a febrile tracheobronchitis, dyspnea, and pain in the elbows and

knees. The etiology of nausea and vomiting is not clear but could be related to the absorption of bowel gas owing to the absence of nitrogen.¹⁷

Caldwell et al⁸⁵ studied four subjects exposed to P_{iO_2} 0.98 atm for, respectively, 30, 48, 60, and 74 hours (i.e., 1.25, 2.0, 2.5, and 3.1 days). They were instructed to take five deep breaths and a cough every 2 hours throughout the day but not at night. In addition to chest pain and cough, three of the four had paresthesia, and three had anorexia. They noted decreases in lung volumes, diffusing capacity, but not in the ratio of forced expiratory volume in 1 second to forced vital capacity, and a modest increase in alveolar-arterial (A-a) PO_2 gradient (89 Torr). Chest radiographs were clear. The authors could not exclude atelectasis as the etiology of the decreased vital capacity. The subject with the longest (74 hours) exposure, however, was hospitalized for 2 days postexposure “until his sense of well-being and appetite returned.” This subject had dyspnea on exertion for 3 to 4 days postexposure, and his vital capacity did not return to baseline until 24 weeks postexposure.⁸⁵ In humans, normobaric hyperoxia does not produce airway obstruction,⁷¹ but may do so in dogs^{11,86} and primates.⁸⁷ Hyperbaric hyperoxia may produce airway obstruction in humans.²⁵

The longest voluntary human exposures to 100% O_2 are by Dolezal.⁸⁸ Twelve subjects were exposed to the limit of tolerance. The mean exposure was 74 hours (3.1 days), with a range from 42 hours (1.75 days) to a remarkable 110 hours (4.6 days). P_{CO_2} of the exposure chamber remained high, approximately 7 to 8 torr (1%). In addition to retrosternal pain and cough, the subjects eventually experienced dyspnea, loss of gustatory sensation, anorexia, nausea (after 48 to 60 hours) and vomiting, general weakness, and vertigo (beyond 60 to 72 hours). Three developed fevers of 38.5°C to 39.3°C (101.3°F to 102.7°F). As evidence of additional systemic (nonrespiratory) toxicity, nine of the twelve noted paresthesias and hypesthesias of the finger tips in both hands and feet, which did not resolve until 14 to 21 days postexposure. Vital capacity decreased by 17%, minute ventilation progressively increased by 39% throughout the exposure, and a mild respiratory alkalosis developed. Remarkably, arterial O_2 saturation remained at or near 100% throughout the exposure and was normal postexposure. Dolezal interpreted this intact oxygenation as evidence that pulmonary edema and “hepatization foci,” as seen in animal studies, had not developed.⁸⁸

Taken together, these reports seem to imply that evidence for major sustained parenchymal lung injury, or the equivalent of acute respiratory distress syndrome (ARDS), has been much more readily observed in animals than in normal humans.

Toxicity of Oxygen Near 1 Atmosphere in Patients

Most patients who receive prolonged high have severe parenchymal lung injury (e.g., ARDS). Intriguing data are available, however, from human patients without major lung disease. In eighteen patients postoperative from cardiac surgery, Singer et al⁸⁹ confirmed that short exposures to 100% O_2 for a mean of 24 hours (range 15 to 48 hours) have no demonstrable adverse effects. They found no change in shunt, compliance, dead space, radiologic atelectasis, or clinical course compared to a group who received only the FI_{O_2} (mean: 0.32) required for adequate oxygenation. Singer et al⁸⁹ also reported two additional patients who were ventilated with 100% O_2 for 5 and 7 days, respectively. Throughout the 5-day and 7-day periods, these patients showed progressive improvement in Pa_{O_2} , from 368 to 430 torr and from 155 to 404 torr, respectively. The first patient was extubated on the fifth day and subsequently discharged from the hospital. The second patient died on the seventh day of massive pulmonary hemorrhage related to a greatly prolonged bleeding time on anticoagulants.

In a companion paper by the same group, Barber et al⁹⁰ reported the effects of ventilating patients with massive cerebral trauma and resultant irreversible, ultimately fatal brain damage, with either 100% O_2 ($n = 5$) or air ($n = 5$). Ventilation with the respective gases continued until death, which occurred within 31 to 72 hours, or 1.3 to 3 days (mean: 2.2 days), in the O_2 group and within 50 to 216 hours, or 2.1 to 9 days (mean: 4.3 days), in the air group. None had preexisting lung disease. None had spontaneous respirations. All received tidal volumes greater than 800 mL, periodic tracheal suctioning, frequent changes in body position, and treatment with corticosteroids. After 2 days, the O_2 group had a lower mean Pa_{O_2} (120 vs. >400 torr, tested on 100% O_2 in both groups), greater shunt and dead space, but no difference in compliance. The O_2 group also had greater worsening of chest radiographs, progressing to multilobar infiltrates, consolidation, or collapse. At autopsy the O_2 group also had greater lung weights. Perhaps the most striking

finding was that at autopsy no histologic differences between the groups were apparent. Both had varying degrees of bronchopneumonia, intraalveolar and interstitial edema, atelectasis, congestion, hemorrhage, and intravascular coagulation. Hyaline membranes, however, were “conspicuously absent” in both groups. It seems likely that hyperoxia-induced atelectasis (and perhaps ventilator-induced lung injury, with an 800-mL tidal volume), could explain many if not all of their findings, especially in such patients with no spontaneous respiratory effort.

Hyde and Rawson⁹¹ reported the effects of inadvertent exposure to 83% to 91% O₂ for 12 to 32 days in five intubated patients with neuromuscular disease. The authors made a presumptive diagnosis of hyperoxic pneumonitis based on scattered patchy infiltrates on chest radiographs, fever, and leukocytosis. The postmortem examination of the one patient who died (a depressed patient who was found dead with the respirator disconnected) showed “thickening of the alveolar septa by edema and fibroblastic proliferation ... fibrin deposition and increased numbers of lining cells in the alveolar lumina.” No hyaline membranes were described. However, interpretation of their findings is confounded by the fact that these patients also had copious secretions, “a major degree of atelectasis” (four of the five patients), and pneumonia (four of five), which are common complications of such patients even in the absence of hyperoxia. Furthermore, unlike ARDS, these patients responded rapidly to minimizing the F_IO₂, “vigorous tracheal aspiration (and) meticulous tracheal toilet,” and antibiotics. In fact, after these changes in management, all four were weaned from mechanical ventilation within 2 to 5 days.

Perhaps the longest reported exposure of a patient to hyperoxia was that of a 32-year-old man with myasthenia gravis, who was ventilated with 80% O₂ (which could not be lowered for technical reasons) for 150 days, at which time he developed blindness bilaterally with retinal artery constriction and retinal atrophy.⁹² Hyperoxia constricts retinal arteries in adults as well as neonates.⁹³ During the exposure his P_aO₂ remained 250 to 300 torr. He remained another 130 days on 60% O₂ (P_aO₂ 120 to 160 torr) before weaning from mechanical ventilation on day 280, after which his P_aO₂ remained normal (80 to 100 torr). Except for the elevated P(a-a)O₂ gradient, consistent with atelectasis, inadequate control of secretions, or pneumonia, no signs or symptoms of pulmonary O₂ toxicity were reported.

In a brief communication Smith et al⁹⁴ reported the absence of any evidence of O₂ toxicity, as determined by radiographs, pulmonary function, bronchoscopy, or histopathology, in forty-one patients treated with high-frequency jet ventilation using a mean F_IO₂ of 0.92 (range: 0.80 to 0.95) for a mean of 4.1 days (range: 8 hours to 12 days). They suggested that their lack of O₂ toxicity may be secondary to the lack of “stretching and shearing forces” encountered in conventional mechanical ventilation.

More recently, Capellier et al⁹⁵ retrospectively studied seventy-four patients with F_IO₂ ≥0.90 continuously for at least 48 hours. They found that the duration of this exposure in the seventeen survivors was surprisingly long (mean: 5.6 days) and not different from the fifty-seven nonsurvivors (5.9 days). One survivor had exposure to F_IO₂ ≥0.90 for 15 days and was eventually discharged, breathing spontaneously with a normal P_aO₂ on room air. The majority of deaths were related to sepsis and multiorgan failure rather than to progressive hypoxemic respiratory failure. Of the thirty-seven patients exposed to F_IO₂ ≥0.90 for at least 4 days, only 5 (14%) died with hypoxemia. These authors concluded that the lungs of patients with acute respiratory failure appear relatively resistant to prolonged hyperoxia, that high plateau pressures may be more injurious than high F_IO₂, and that a prospective trial may be helpful in clarifying the optimum management of F_IO₂ for such patients.⁹⁵

Pathology of Pulmonary Oxygen Toxicity: Diffuse Alveolar Damage

In 1958 Pratt⁹⁶ was the first to note in patients pathologic changes attributed to hyperoxia. The pathology of pulmonary O₂ toxicity has been reviewed by a number of authors^{13,16,20,23-26,28,97,98} and is the subject of a monograph.¹¹ Because of preexisting lung disease in most human studies, the most persuasive data come from animals. A few studies, however, of hyperoxic pulmonary pathology in humans without preexisting lung disease are available,⁹⁹⁻¹⁰¹ and the findings are consistent those in animals. The histologic pathology of hyperoxia-induced acute lung injury is that of diffuse alveolar damage.⁹⁸ *Diffuse alveolar damage* is a descriptive term for the

predictable but nonspecific features of acute lung injury from a variety of toxins, including infectious agents, other inhalants, pharmaceutical agents and other ingestants, radiation, and the multiple causes of ARDS, including trauma and sepsis.⁹⁸ Hyperoxia has in common with all other causes of diffuse alveolar damage an initial injury to both the alveolar endothelial and epithelial cells.

Diffuse alveolar damage can be divided into two stages, as first defined by Nash et al,¹⁰² and detailed in Table 45-2.⁹⁷ The first is an early acute or exudative stage within the first week, characterized by interstitial and intraalveolar edema, intraalveolar hemorrhage and fibrin deposition, sloughing of the alveolar lining cells (type I pneumocytes) with denudation of the alveolar basement membrane, and hyaline membranes, which represent cytoplasmic and nuclear debris from sloughed cells mixed with fibrin.⁹⁸ Late in the first week an inflammatory cell infiltrate into the interstitium is evident. The small pulmonary arterioles and alveolar capillaries may show fibrin thrombi. Also by the end of the first week, proliferation of type II pneumocytes appears along the alveolar lining, representing a reparative phase; the type II pneumocytes can later differentiate into type I pneumocytes.⁹⁸

Table 45-2: Diffuse Alveolar Damage in Humans at 0.8 to 1 Atmospheres of Oxygen

Changes	Time of Occurrence
A. Exudative	
1. Capillary congestion, focal intraalveolar edema, few cases with fibrin thrombi in capillaries and small arteries	Less than 3 days
2. Interstitial edema	After 3 days
3. Hyaline membranes lining alveolar septa	22 hours to 7 days, decreasing or focal thereafter
4. Interstitial mononuclear cell infiltrate (lymphocytes, plasma cells, unclassified cells)	Less than 1 week
B. Proliferative	
1. Alveolar lining cell hyperplasia	Focal at 3 days, diffuse at 1 week
2. Interstitial fibrosis	Focal at 3 days, diffuse at 8 days, severe by 2 weeks

Derived from light microscopic data of Katzenstein et al.¹⁰⁰

Data from Balentine JD. *Pathology of Oxygen Toxicity*. New York: Academic Press; 1982.

The second stage of diffuse alveolar damage⁹⁸ is a proliferative or organizing stage and is noted beyond the first week. Fibroblastic proliferation in the interstitium and focal intraalveolar fibrosis appear. Whereas the edema and hyaline membranes have largely resolved, the interstitial inflammatory infiltrate and the alveolar lining cell hyperplasia are still present. A striking interstitial fibrosis develops in association with extensive interstitial deposition of collagen.

The issue of just how sensitive the human lung is to hyperoxic lung injury is still somewhat open to question. Remarkably, as noted above, in patients without prior lung injury who received 100% O₂ for up to 7 days, typical histologic lesions were not found.^{89,90} These observations prompted Katzenstein⁹⁸ to suggest that humans may be susceptible to significant hyperoxic lung injury only after their lungs are first damaged by other insults. Other authors have concluded simply that no consistent evidence shows that hyperoxia (F_IO₂ >0.60) is dangerous in acute lung injury or ARDS.¹⁰³

Bronchopulmonary Dysplasia

Bronchopulmonary dysplasia was initially described in 1967¹⁰⁴ and has been reviewed in detail by Balentine¹¹ and Truog.¹⁰⁵ It is a form of chronic lung disease that follows therapy for idiopathic respiratory distress syndrome of the newborn, or hyaline membrane disease. Bronchopulmonary dysplasia develops in very low-birth-weight infants (≤ 1500 g at birth). Its incidence and prevalence have risen substantially in recent years because of the markedly increased survival of these infants.¹⁰⁵ It has become the most common cause of chronic lung disease in infants.¹⁰⁶ Its pathologic features include fibrosis and destruction of acinar structures with a resultant combination of scarring and emphysematous changes. It appears to have three contributing mechanisms: (a) ventilator-induced lung injury,¹⁰⁷ (b) hyperoxia, with increased O_2 radicals in the presence of an immature antioxidant defense system,¹⁰⁸ and (c) peribronchial pulmonary edema secondary to the birth-related increased pulmonary blood flow through an immature and injured pulmonary microcirculation.¹⁰⁵ Because high levels of both mechanical ventilation and hyperoxia virtually always occur together in patients with idiopathic respiratory distress syndrome, the relative roles of these two etiologic factors is unclear. Churg et al¹⁰⁹ have described the equivalent of bronchopulmonary dysplasia, including “honeycombing,” in three adult patients who also had received the combination of high ventilator pressures and high FI_{O_2} for 3 to 7 weeks. Baboon and other animal models for bronchopulmonary dysplasia exist.^{110,111}

In recent years, the risk and severity of bronchopulmonary dysplasia have decreased with the use of antenatal glucocorticoids, less aggressive mechanical ventilation, including limiting the FI_{O_2} and the use of continuous positive airway pressure,^{112,113} and postnatal surfactant.^{112–114}

Nonrespiratory Effects of Oxygen Breathing

The nonrespiratory effects of hyperoxia have been extensively reviewed elsewhere.^{11,13,28,115} The best-recognized effects include hemodynamic changes, suppression of erythropoiesis and serum erythropoietin,¹¹⁶ the Haldane effect (see the section [Respiratory Depression and Stimulation, Pulmonary Vasodilation, and Hypercapnia](#)), retinopathy of prematurity (previously called retrolental fibroplasia), and, only at O_2 pressures exceeding 2 atm, seizures.

Retinopathy of Prematurity

Retinopathy of prematurity results from hyperoxic vasoconstriction and injury to the exquisitely susceptible, growing retinal capillaries of premature neonates (<26 weeks gestation).^{11,117,118} The resulting paradoxical “hyperoxic hypoxia” and tissue ischemia induces neovascularization within the retina and adjacent vitreous. This retinovitreal neovascularization eventually results in retinal detachment and blindness,¹¹⁸ a process similar to that in diabetic and sickle-cell retinopathies.¹¹⁷ Both the incidence and severity of retinopathy of prematurity correlate with the duration of exposure to Pa_{O_2} greater than 80 torr.¹¹⁹ The pathogenesis of retinopathy of the newborn has two phases.¹²⁰ The first phase (20 to 30 weeks postmenstrual age) involves hyperoxia and decreased vascular endothelial growth factor (VEGF) levels, whereas the second phase (31 to 44 weeks) involves hypoxia and increased VEGF levels. Activation of the VEGF receptor VEGFR-1 can prevent the hyperoxia-induced retinal vascular degeneration of retinopathy of prematurity.¹²¹ More recently, it has been reported that intravitreal bevacizumab (anti-VEGF) benefitted infants with retinopathy of prematurity.¹²⁰ The optimum oxygenation is unclear. The recent SUPPORT trial in premature infants reported that a lower target range of arterial oxyhemoglobin saturation (Sp_{O_2}) (85% to 89% vs. 91% to 95%) increased mortality but decreased severe retinopathy in the survivors.^{112,113}

Acute Hemodynamic Effects

The acute hemodynamic effects of normobaric hyperoxia in healthy humans have been studied by a number of investigators^{115,122–134} and summarized by Lodato,^{28,115,130} who also found virtually identical responses in conscious dogs.^{115,130} [Table 45-3](#) summarizes these results. The decrease in heart rate is vagally mediated.^{33,123} The decrease in cardiac output is primarily the result of

the relative bradycardia, but some studies also have shown an independent contribution from a decrease in stroke volume or diastolic dysfunction.¹³⁵ The decrease in cardiac work¹¹⁵ is primarily the result of the bradycardia, but right-ventricular afterload is reduced by pulmonary vasodilation.¹³⁶

Table 45-3: Acute Hemodynamic Effects of Oxygen Breathing (Normobaric Hyperoxia)

Parameter	Effect
Heart rate	Decreased
Cardiac output	Decreased
Right-ventricular stroke work	Decreased
Left-ventricular stroke work	Unchanged
Right- and left-ventricular work rates	Decreased
Systemic arterial pressure	Variably increased, decreased, or unchanged
Pulmonary arterial pressure	Decreased
Pulmonary arterial wedge pressure	Unchanged
Right-atrial pressure	Increased
Systemic vascular resistance	Increased
Pulmonary vascular resistance	Decreased

Data from Lodato.¹¹⁵

Systemic vasoconstriction is one of the most consistent findings and occurs independently of any changes in P_{CO_2} .^{137,138} Virtually all systemic beds constrict during hyperoxia, but in rats the splanchnic circulation vasodilates.¹³⁹ The central nervous system vasoconstrictive effect has been used to treat vascular headaches.¹⁷ The mechanisms are not fully understood and appear quite diverse among various vascular beds. Hyperoxic vasoconstriction may be a continuum of the mechanism of hypoxic vasodilation.¹⁴⁰ That is, red blood cells can regulate vasomotor tone depending on their degree of oxyhemoglobin saturation: As saturation rises with hyperoxia (or falls with hypoxia), the red blood cell releases less (or more, respectively) of the vasodilators adenosine 5' triphosphate¹⁴¹ and nitric oxide from S-nitroso-hemoglobin.¹⁴² In the central nervous system^{143,144} and in the human forearm,¹⁴⁵ hyperoxia vasoconstricts by inactivation of endothelium-derived relaxing factor/nitric oxide by superoxide anion.^{140,146} In the coronary bed, hyperoxia constricts by closure of adenosine 5' triphosphate-sensitive potassium channels.¹⁴⁷ In the human retinal artery the reported mechanisms include release of endothelin-1¹⁴⁸ and prostanoids,¹⁴⁹ in addition to the mechanisms above.¹³⁸ The human umbilical vein vasoconstricts to hyperoxia via local release of norepinephrine.¹⁵⁰

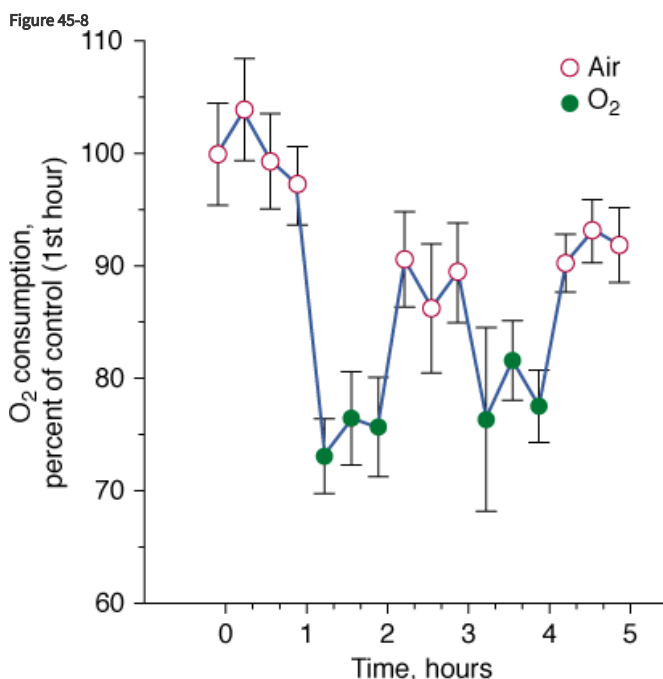
Despite the systemic vasoconstriction, systemic arterial pressure is variably affected, owing to the concurrent decrease in cardiac output. Jubran and Lodato¹⁵¹ showed that the bradycardia is independent of the arterial baroreceptor reflex.

The hemodynamic changes induced by hyperoxia have also been documented in patients with a wide variety of conditions, including general anesthesia,¹⁵² during cardiopulmonary bypass,¹⁵³ immediately following coronary bypass surgery,¹⁵⁴ with cardiac risk,¹⁵⁵ congestive heart failure,^{135,156,157} sepsis,^{158,159} cirrhosis,¹⁶⁰ and children with acyanotic congenital heart disease.¹⁶¹

In patients with pulmonary hypertension from a variety of causes (primary, portal, human immunodeficiency virus, atrial septal defect, ventricular septal defect, scleroderma) with mean baseline $P_{aO_2} = 64$ mm Hg, breathing 100% O_2 has been reported to decrease pulmonary vascular resistance by 24%.¹⁶² A limitation of this study is that cardiac output was calculated from the Fick principle, assuming that hyperoxia did not change O_2 consumption, an invalid assumption.¹¹⁵

Oxygen Consumption

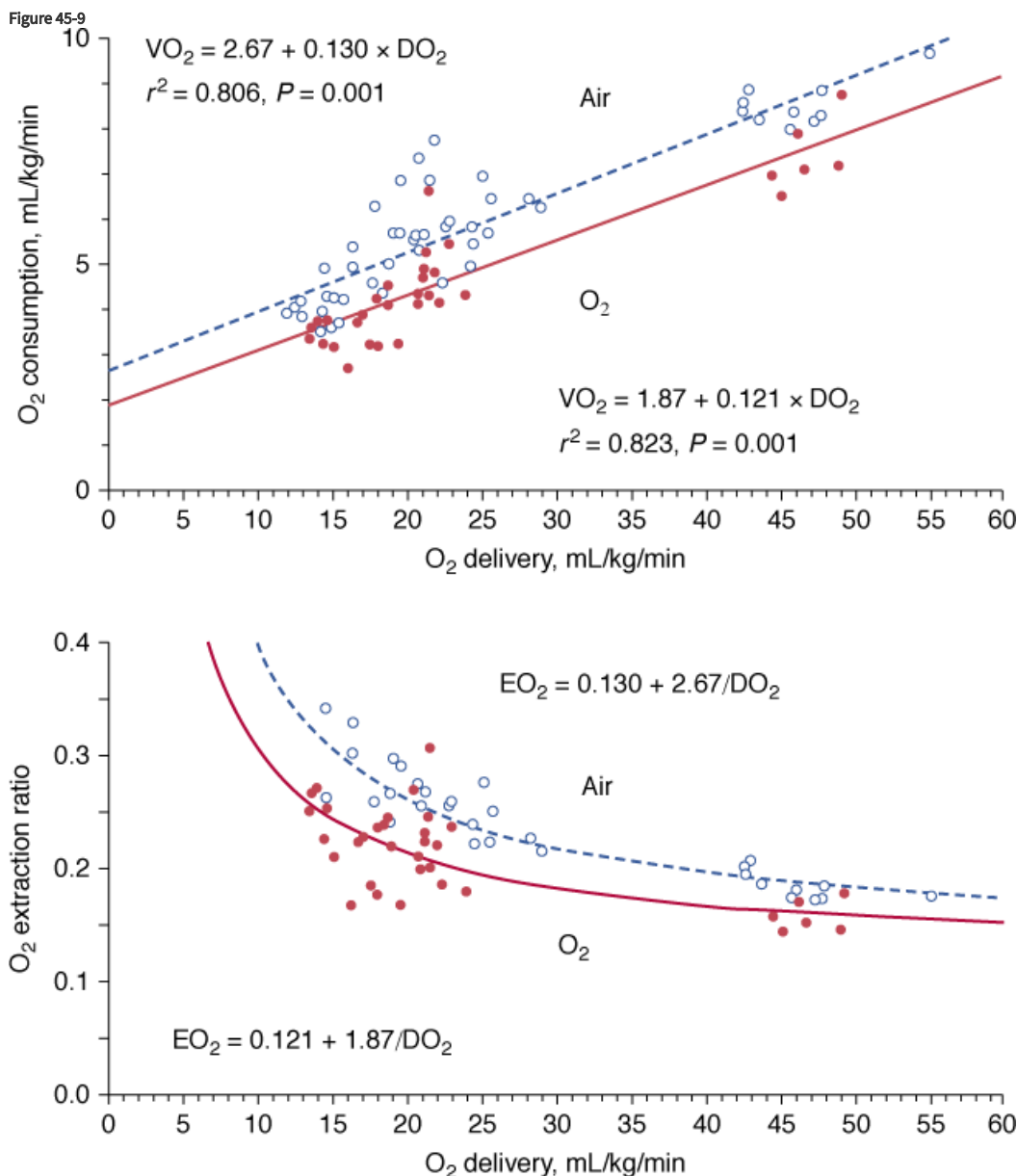
Contrary to Priestley's speculation, which he based on his observations of a candle burning more vigorously in O_2 (see above), that hyperoxia may enhance metabolic rate, recent evidence indicates that hyperoxia can actually decrease metabolic rate as measured by O_2 consumption. It has long been established that hyperoxia decreases O_2 consumption in both cells and tissue preparations, but the biochemical mechanisms are unclear.¹⁶³ Chapler et al¹⁶⁴ showed that under certain experimental conditions (e.g., acute anemia, autonomic inhibition) in anesthetized dogs, hyperoxia paradoxically decreased O_2 consumption. Lodato¹¹⁵ showed under more general physiologic conditions in unsedated, intact, healthy, conscious dogs, that hyperoxia (mean $P_{aO_2} = 475$ torr) decreased whole-body O_2 consumption by nearly 20% (Fig. 45-8). Unexpectedly, in this study the arteriovenous difference in O_2 content decreased along with the decrease in cardiac output (both by approximately 10%). Paradoxically, O_2 delivery (cardiac output $\times$ arterial O_2 content) did not change, and O_2 extraction (O_2 consumption/ O_2 delivery) decreased. Subsequently, Lodato¹³⁰ reported that when O_2 consumption and O_2 delivery were plotted in the conventional way, as ordinate and abscissa, respectively, hyperoxia produced a unique parallel shift downward in the O_2 consumption– O_2 delivery relationship (Fig. 45-9). Hyperoxia-induced decreases in O_2 consumption have also been reported in children with acyanotic congenital heart disease,¹⁶¹ patients with sepsis,^{158,159} cardiac-risk patients,¹⁵⁵ and patients during cardiopulmonary bypass.¹⁵³



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Time course of resting whole-body O_2 consumption as the inspired gas was alternated hourly between air (open circles) and O_2 breathing (hyperoxia; mean $P_{aO_2} = 475$ torr) (closed circles) in six conscious dogs. Each point is expressed as percent of mean O_2

consumption ($6.19 \pm 0.65 \text{ mL O}_2 \cdot \text{min}^{-1} \text{kg}^{-1}$) during control period (first hour). Paradoxical decrease in O_2 consumption during hyperoxia was fully developed by 20 minutes, was maintained for at least 1 hour, and was both reversible and reproducible. Mean of each 1-hour period was statistically significantly different from mean of its adjacent 1-hour periods. (Used, with permission, from Lodato.¹¹⁵)



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Upper panel. Relationship between O_2 consumption and O_2 delivery in six healthy, resting, conscious dogs for air (open circles) versus O_2 breathing (normobaric hyperoxia; closed circles). Each dog is represented by $n = 10$ independent determinations during air breathing and by $n = 6$ independent determinations during hyperoxia. Hyperoxia produced a unique parallel shift downward in the O_2 consumption– O_2 delivery relationship (see text). **Lower panel.** Oxygen extraction ratios corresponding to the data points in upper panel. (Used, with permission, from Lodato.¹³⁰)

The mechanisms by which hyperoxia could induce such a decrease in O_2 consumption are obscure. The possibilities include¹¹⁵ that hyperoxia may produce (a) systemic cellular O_2 toxicity, (b) a paradoxically inadequate O_2 supply (“hyperoxic hypoxia”) at the microcirculatory level, owing to the diffuse systemic vasoconstriction,^{165–168} or (c) a facultative decrease in O_2 demand, as has been

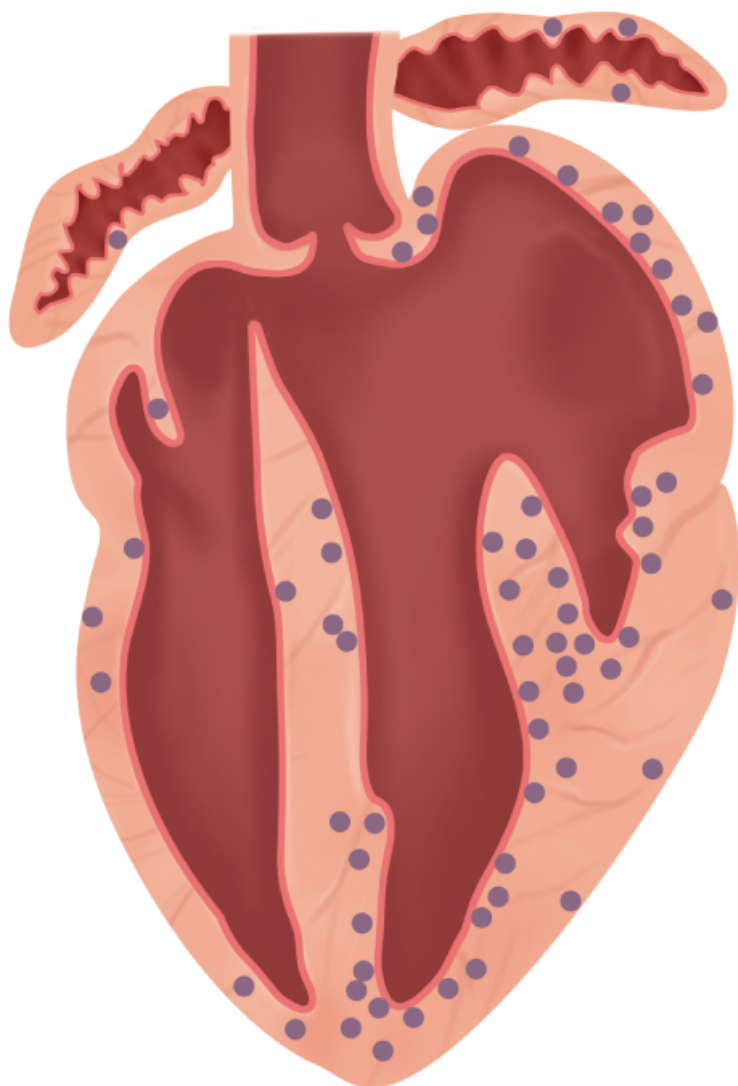
demonstrated in mammals under certain physiologic challenges unrelated to hyperoxia.¹⁶⁹ Reinhart et al¹⁵⁹ investigated the second possibility. In patients with sepsis¹⁵⁹ and in stable patients with cardiac risk,¹⁵⁵ they infused the antioxidant *N*-acetylcysteine. They found that *N*-acetylcysteine attenuated the hyperoxia-induced decreases in both O₂ consumption (from 34% to only 11%) and gastric intramucosal pH (by tonometry, used as a marker of the adequacy of gastric O₂ supply).¹⁵⁹

Pulmonary versus Systemic Cause of Death from Hyperoxia

Death in experimental animals exposed to prolonged hyperoxia is usually attributable to ARDS-like acute lung injury.^{170,171} Hypoxemia, however, is generally absent. Matalon et al¹⁷¹ found in sheep a terminal severe acute respiratory acidosis but supranormal Pa_{O_2} = 200 torr at death. Harabin et al¹⁷² provided intriguing evidence that *systemic*, rather than pulmonary, O₂ toxicity may be the predominant cause of death. Dogs exposed continuously to 100% O₂ lived for a mean of 88 hours (3.7 days), but surprisingly, they maintained Pa_{O_2} at approximately 500 torr until shortly before death. Their brief preterminal course was characterized by a precipitous decrease in cardiac output and blood pressure accompanied by metabolic acidosis, all of which preceded the terminal deterioration in gas exchange. In fact, the acidosis (presumably lactic as seen terminally in hyperoxemic rabbits¹⁷³) occurred in the face of supranormal Pa_{O_2} of 200 to 400 torr, indicating a systemic, or cardiovascular, rather than a pulmonary (hypoxemic) mechanism of their terminal deterioration.¹⁷² Thus, these animals died clearly *with* lung injury but not so clearly *from* lung injury.

Further evidence of primary cardiovascular O₂ toxicity has been provided by Busing et al¹⁷⁴ (and others¹¹), who described disseminated microscopic foci of myocardial necrosis (Fig. 45-10), especially in the subendocardium of the left ventricle, occurring in rabbits subjected to 60 hours (2.5 days) of 100% O₂. Hypoxemia from hyperoxic lung injury was excluded as the mechanism because the cardiac necrosis developed while the Pa_{O_2} was 300 to 400 torr, and, in fact, it preceded any morphologic pulmonary changes. Fracica et al^{87,175} reported that baboons exposed to 100% O₂ for more than 80 hours (>3.3 days) had substantial deterioration in cardiac function without significant deterioration in oxygenation (mean Pa_{O_2} = 437 torr). These animals also showed focal areas of myocardial necrosis.¹⁷⁵ Robinson et al¹⁷⁶ studied several species of primates exposed to 100% O₂ for up to 14 days. Pa_{O_2} remained high (400 to 500 torr) until near death when it fell sharply and Pa_{CO_2} rose. The baboons, however, did have other clinical (increase respiratory effort) and pathologic evidence of severe lung injury. In contrast, the squirrel monkeys were surprisingly insensitive to pulmonary O₂ toxicity. They, too, died from the hyperoxic environment but of undetermined cause. Cardiac histopathologic examination was not reported.

Figure 45-10



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Distribution of disseminated microscopic foci of myocardial necrosis seen in rabbits exposed to 100% O₂ (normobaric hyperoxia) for more than 60 hours, as illustrated by a schematic longitudinal section through all chambers of the heart. All four chambers were affected, but the subendocardium of the left ventricle, including the papillary muscle and the septum, were the most vulnerable. The arterial O₂ tension was consistently greater than 300 torr, indicating that hypoxemia (from possible hyperoxic lung injury) was not the mechanism of the necrotizing cardiomyopathy. (From Busing CM, Kreinsen U, Buhler F, Bleyl U. Light and electron microscopic examinations of experimentally produced heart muscle necroses following normobaric hyperoxia. *Virchows Arch A Pathol Anat Histol.* 1975;366:137–147. With kind permission from Springer Science and Business Media.)

Thus, it appears that experimental animals exposed to hyperoxia die primarily from systemic rather than pulmonary O₂ toxicity.

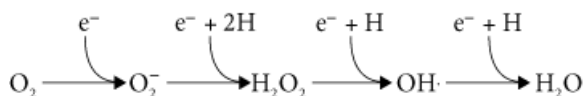
Mechanisms of Oxygen Toxicity

Formation of Reactive Oxygen Species

It is now generally accepted that the toxic effects of hyperoxia are the direct result of increased concentrations of highly reactive O₂-derived free radicals.^{20,177–181} This extensive topic is now the subject of two recent monographs.^{182,183} The “O₂-radical hypothesis” was first proposed in 1954 by Gerschman et al,¹⁸⁴ who suggested that hyperoxia-induced tissue injury and X-irradiation-induced tissue injury have as a common mechanism the production of reactive O₂ species in excess of the antioxidant defenses. Gerschman¹⁸⁵

pointed out that O_2 at 20% is potentially toxic and that its gradual accumulation in the atmosphere induced the evolution of cellular defense mechanisms.

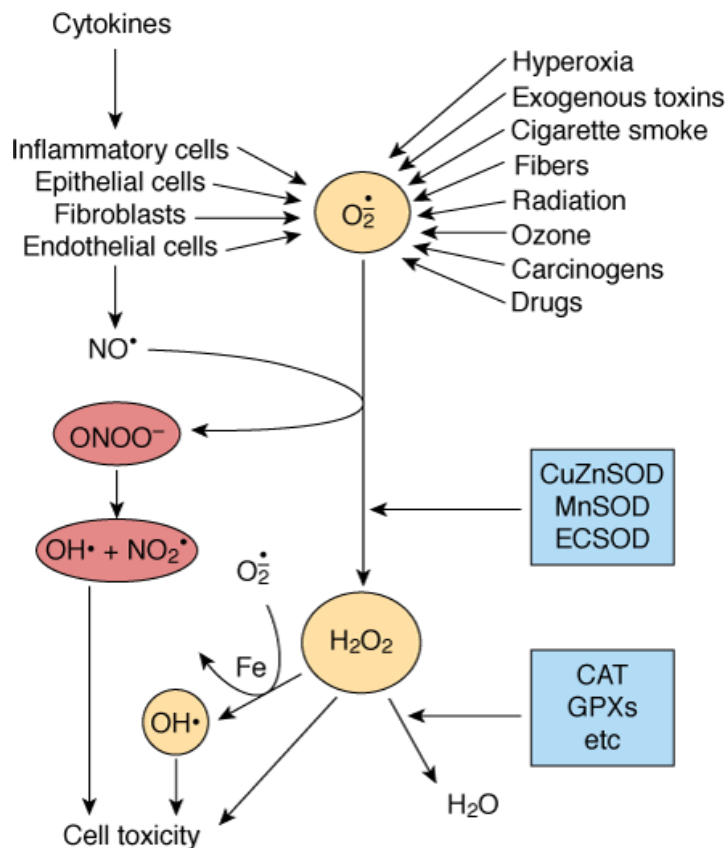
The basic chemistry of reactive O_2 species has been extensively reviewed.^{186–190} Molecular O_2 itself is generally nontoxic and only modestly reactive. When it is chemically reduced by the sequential addition of electrons (e^-), however, it forms partially reduced reactive O_2 species (ROS), as follows:



where O_2^- is superoxide anion radical, H_2O_2 is hydrogen peroxide, and $OH^\cdot$ is the hydroxyl radical. Superoxide is only modestly reactive. It can, however, yield the very highly reactive hydroxyl radical. Alternatively, superoxide can react with [nitric oxide](#) (NO), also a free radical, forming the very highly reactive peroxynitrite anion, $ONOO^-$. Of the ROS, the hydroxyl radical is by far the most reactive, reacting with all known biomolecules so rapidly that it acts only locally where it is produced.¹⁸⁷ In contrast O_2^- and H_2O_2 can travel some distance and enter cells, H_2O_2 by simple diffusion across the cell membrane and O_2^- via anion channels.^{189,191} The high reactivities of ROS and reactive nitrogen species (RNS) such as NO and peroxynitrite are the result of their very high affinity for additional electrons. This extreme electron affinity causes these agents to rapidly pull an electron from the nearest available molecule, resulting in oxidative damage to nearby lipids, proteins, and DNA. This oxidant damage is manifested by lipid peroxidation, enzyme inhibition, and DNA strand breakage, which can lead ultimately to loss of cell integrity and cell death.

The sources of ROS are many ([Fig. 45-11](#)). In the normal processes of cellular metabolism, ROS are produced at basal rates. Superoxide radical can be generated by both enzymatic and nonenzymatic (autooxidation) reactions. The major source of ROS in the lung is the mitochondrial electron-transport chain, during the production of adenosine 5' triphosphate via oxidative phosphorylation.¹⁹⁰ The fate of nearly all O_2 taken up by the body is four-electron catalytic reduction by mitochondrial cytochrome c oxidase to form water. Normally, a tiny fraction (0.1% to 2%^{189,192}) of the electron flow along this chain “leaks” off upstream of cytochrome oxidase, principally at the reduced form of nicotinamide adenine dinucleotide (NADH) dehydrogenase and ubiquinone ([Fig. 45-12](#)), and partially reduces O_2 to yield O_2^- .^{178,193} The other major source of intracellular superoxide generation is the plasma membrane nicotinamide adenine dinucleotide phosphate (NADPH) oxidase.^{163,194}

Figure 45-11



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Sources of reactive oxygen species (ROS) and reactive nitrogen species (RNS) in the lung. Both inflammation (*upper left*) and exogenous toxins (*upper right*), including hyperoxia, produce superoxide ($O_2^{\bullet -}$), which can react with nitric oxide (NO) to produce peroxynitrite ($ONOO^-$) and other toxic RNS and ROS. Or, superoxide may be metabolized to hydrogen peroxide (H_2O_2), which may yield water (H_2O)

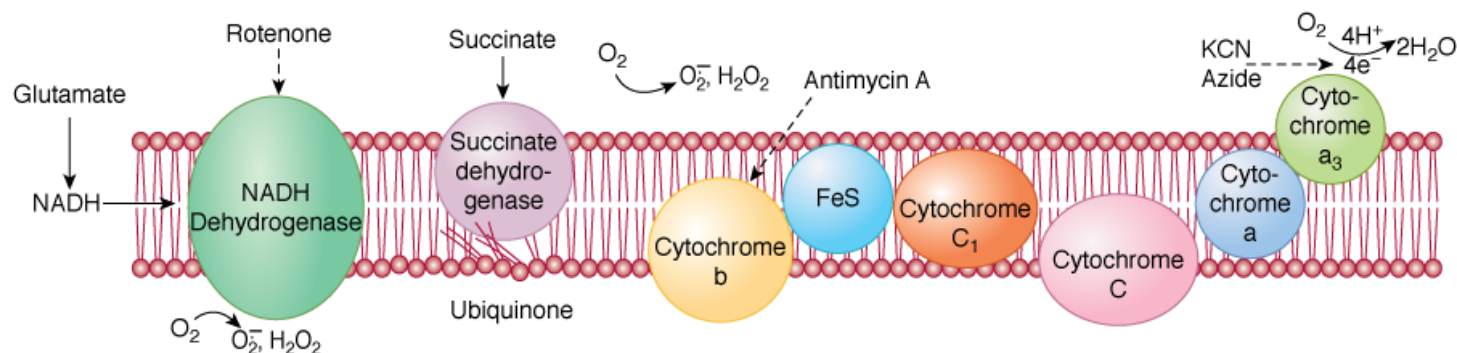
or the toxic hydroxyl radical ($OH^\bullet$). See text for details. CAT, catalase, GPx, glutathione peroxidase, SOD, superoxide dismutase.

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Superoxide dismutases in the lung and human lung diseases. *Am J Respir Crit Care Med*. 2003;167(12):1600–1619. Official Journal of the American Thoracic Society.)

Figure 45-12

Mitochondrial electron transport



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

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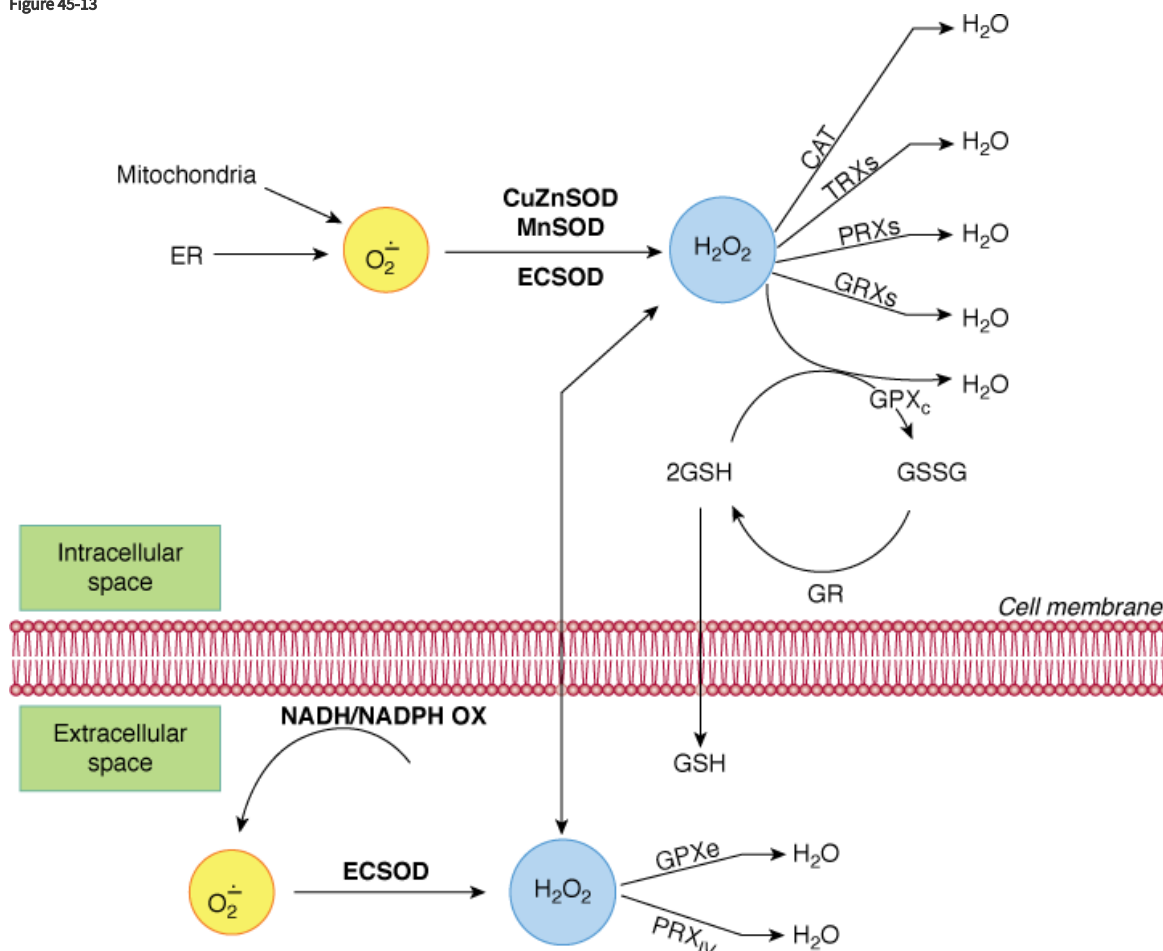
Generation of reactive oxygen species (ROS) in the mitochondria. The major source of mitochondrial ROS is the electron transport chain located on the inner mitochondrial membrane. Ubiquinone is reduced in a one-electron transfer to form the ubisemiquinone radical, which is reoxidized by molecular oxygen (autooxidation) to form the superoxide radical ($O_2^{\bullet -}$); this reaction is the major source of

mitochondrial superoxide radical, which serves as a precursor to hydrogen peroxide (H_2O_2) and the hydroxyl radical ($\text{OH}^\cdot$), as shown. The autooxidation of the reduced form of nicotinamide adenine dinucleotide (NADH) dehydrogenase (a member of the class of flavoproteins) by molecular O_2 is an additional but quantitatively less important source of mitochondrial superoxide radical and thus, hydrogen peroxide and the hydroxyl radical. Also shown are various electron transport inhibitors (*dashed arrows*) used to study mitochondrial sources of superoxide radical. (Reprinted, with permission, from Macmillan Publishers Ltd. From Freeman BA, Crapo JD. Biology of disease. Free radicals and tissue injury. *Lab. Invest.* 1982;47:412–426.)

Under pathologic conditions, such as hyperoxia or inflammation, which may itself result from hyperoxic injury or from invading microorganisms, or other exogenous toxins (ozone, cigarette smoke, fibrogenic material, ionizing radiation, paraquat, or cytotoxic drugs), the production of both ROS and RNS is increased (see Fig. 45-11).^{179,186–190} One hundred percent O_2 results in a 10-fold to 15-fold increase in mitochondrial H_2O_2 production.^{195,196}

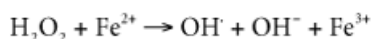
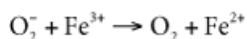
Inflammation results in increased production of ROS and RNS by release of proinflammatory chemokines and cytokines, which recruit neutrophils and other inflammatory cells (see Fig. 45-11). These proinflammatory stimuli also activate the membrane-bound NADPH-oxidase complex, which generates large amounts of superoxide,¹⁸⁹ manifested as the “respiratory burst” of neutrophils and macrophages (Fig. 45-13). Inflammation also upregulates inducible nitric oxide synthase, which produces great quantities of NO from L-arginine.¹⁹⁷ Thus, as above, superoxide and NO may react to yield the very highly reactive peroxynitrite anion, ONOO^- , which may be the most important species responsible for the incapacitation of microorganisms.¹⁸⁷ Inducible nitric oxide synthase can also generate superoxide.¹⁹⁸ An additional enzymatic pathway to increased ROS and RNS, particularly following ischemia, is via xanthine dehydrogenase.¹⁸⁹

Figure 45-13



Major antioxidant pathways in the lung, intracellular and extracellular, scavenging superoxide ($O_2^{\cdot-}$) and hydrogen peroxide (H_2O_2). See text for details. *CAT*, catalase; *ER*, endoplasmic reticulum; *GPXc*, intracellular glutathione peroxidase; *GPXe*, extracellular glutathione peroxidase; *GRXs*, glutaredoxins; *GR*, glutathione reductase; *GSH*, reduced glutathione; *GSSG*, oxidized glutathione; *PRXs*, peroxiredoxins; *SOD*, superoxide dismutase; *TRXs*, thioredoxins. (Reprinted with permission of the American Thoracic Society. Copyright © 2012 American Thoracic Society. From Kinnula VL, Crapo JD. Superoxide dismutases in the lung and human lung diseases. *Am J Respir Crit Care Med*. 2003;167(12):1600–1619. Official Journal of the American Thoracic Society.)

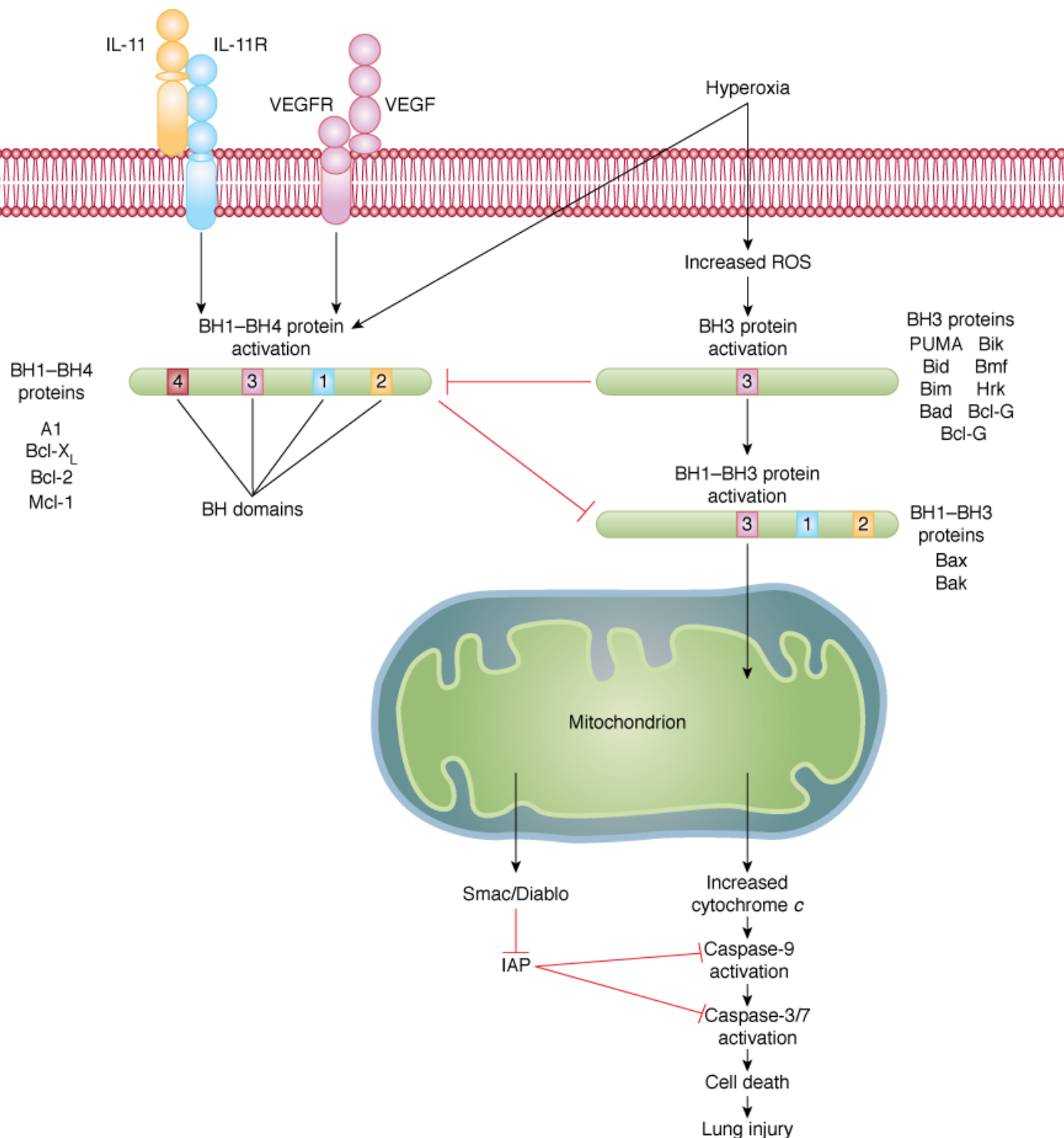
Superoxide radical, produced by any of the several pathways above, is converted to hydrogen peroxide by both spontaneous and enzymatic dismutation. Hydrogen peroxide, in turn, can be further reduced by transition metals to generate the hydroxyl radical. The metal ion-dependent formation of $OH^{\cdot}$ from H_2O_2 is accelerated by $O_2^{\cdot-}$ according to the metal-catalyzed Haber-Weiss reaction, which is referred to as the Fenton reaction if the metal is iron:¹⁹⁹



Cell Death by Hyperoxia: Apoptosis and Necrosis

Recently, attention has focused on the observation that the morphology of alveolar epithelial and endothelial cell death induced by hyperoxia, in vitro and in vivo, is both apoptotic and necrotic.^{200–202} The importance of this observation is that it suggests that the degree hyperoxia-induced lung injury could be modulated by controlling apoptosis,^{202,203} as has been shown in a model of ventilator-induced lung injury.²⁰⁴ Budinger et al.²⁰³ showed in mice that oxidants generated in the mitochondria in response to hyperoxia activate the intrinsic cell death pathway by activating BAX and BAK. When activated, these two proteins create channels in the outer mitochondrial membrane, with release of cytochrome c and other pro-apoptotic proteins from the mitochondrial intermembrane space into the cytosol, leading to activation of the caspases, cell death, and lung injury^{201,203} (see Fig. 45-14). The activity of the pro-apoptotic proteins BAX and BAK are, in turn, counterbalanced by the Bcl-2 family of antiapoptotic proteins, for example, Bcl-2 and A1.²⁰¹ The activity of these antiapoptotic proteins (specifically, A1) can be upregulated by cytokines (e.g., interleukin-11 and VEGF) and even hyperoxia itself^{200,201} (Fig. 45-14), limiting the damage from hyperoxia. These observations provide a basis for potential therapy for hyperoxic lung injury.

Figure 45-14



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Schematic of the intrinsic or mitochondrial apoptosis pathway by which hyperoxia is postulated to cause cell death and lung injury. Cytokines, such as interleukin (IL)-11 and vascular endothelial growth factor (VEGF), and hyperoxia itself, upregulate the activity of A1, a member of the antiapoptotic Bcl-2 family of proteins. A1 downregulates the activity of the proapoptotic proteins BAX and BAK and thereby ameliorates the damage from hyperoxia. See text for details. *IAP*, inhibitors of apoptosis; *IL-11R*, interleukin-11 receptor; *PUMA*, p53 upregulated modulator of apoptosis; *ROS*, reactive oxygen species; *VEGFR*, vascular endothelial growth factor receptor. (Used, with permission, from Budinger and Sznajder.²⁰¹)

Antioxidant Defense Mechanisms

Four antioxidant defense mechanisms exist in the cell²⁰⁵: (a) prevention of the formation of free radicals, (b) conversion of ROS to less-toxic species, (c) compartmentalization of ROS away from vital cellular structures (e.g., the binding of intracellular iron to ferritin to prevent its participation in the Haber-Weiss reaction), and (d) repair of molecular injury induced by ROS.

The primary cellular defense against ROS are enzymes that catalyze the removal of superoxide and hydrogen peroxide, namely, superoxide dismutase (SOD), catalase, and the enzymes of the glutathione redox cycle (see Fig. 45-13).^{186,205} No enzyme-scavenging system exists for the hydroxyl radical.¹⁸⁹ Superoxide radical is converted by dismutation to hydrogen peroxide by SOD:

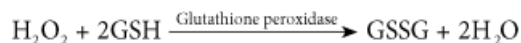


The lung contains three forms of SOD. These have been reviewed in detail.^{186,188} The cytosolic form of SOD contains copper and zinc (CuZnSOD). The mitochondrial form contains manganese (MnSOD). An extracellular form (ECSOD) is concentrated in airway and vessel walls.²⁰⁶

Hydrogen peroxide is converted to water by the enzymes catalase (a hemoprotein) and glutathione peroxidase:¹⁸⁷



Glutathione peroxidase is a selenium-dependent enzyme that inactivates both hydrogen peroxide and lipid peroxides. It is the key enzyme in the glutathione redox cycle (Fig. 45-15),²⁰⁵ and requires the sulfhydryl-containing tripeptide glutathione (GSH; L-γ-glutamyl-L-cysteinyl-glycine) as a cosubstrate. The enzyme catalyzes the reaction that forms oxidized glutathione,²⁰⁷ or glutathione disulfide (GSSG), according to,

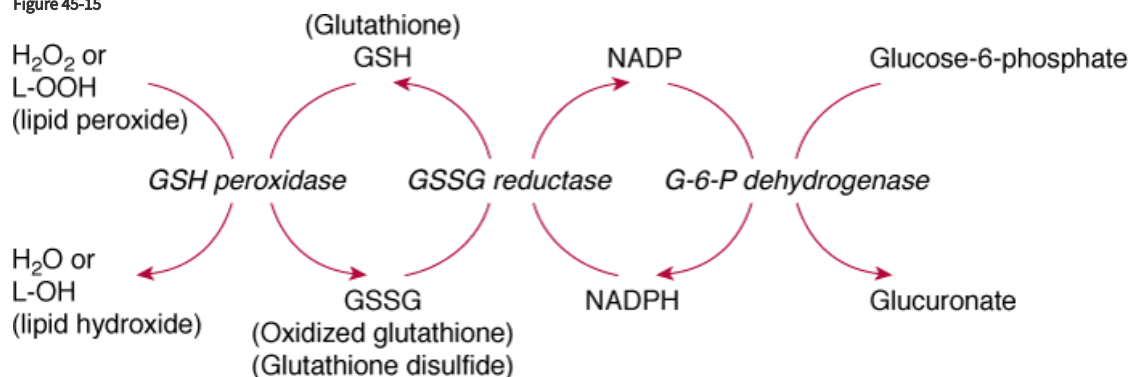


or, in the case of lipid hydroperoxides,²⁰⁵



GSSG is rapidly reduced back to GSH by the NADPH-dependent enzyme, glutathione reductase. NADPH is in turn restored from the oxidized form of nicotinamide adenine dinucleotide phosphate positive (NADP⁺) by the hexose monophosphate shunt (see Fig. 45-12).²⁰⁷

Figure 45-15



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The glutathione (*GSH*) oxidation-reduction (redox) cycle and its pivotal role in the reduction of toxic intracellular hydroperoxides, including hydrogen peroxide and lipid peroxides. The key enzyme in the *GSH* redox cycle is *GSH peroxidase*, a predominantly cytosolic enzyme that requires reduced *GSH* as a substrate. Regeneration of reduced *GSH* by *GSSG reductase*, which is tightly bound to *GSH peroxidase*, requires nicotinamide adenine dinucleotide phosphate (NADPH), which is supplied by *G-6-P dehydrogenase* in the hexose monophosphate shunt.

Within the cell are several other antioxidants.^{186,188} These include water-soluble cytoplasmic antioxidants (e.g., ascorbate [vitamin C]), and fat-soluble membrane-associated antioxidants (e.g., α -tocopherol [vitamin E]), beta-carotene, and ubiquinol. Vitamin E, one of the most important of the nonenzymatic antioxidants, partitions into and protects lipid membranes.²⁰⁵ Vitamin E is particularly effective against $O_2^{\cdot-}$, $OH^{\cdot}$, and blocks propagation of lipid peroxidation by scavenging peroxy free radicals.¹⁸⁸

The human lung expresses several other enzymes capable of inactivating H_2O_2 ,¹⁸⁶ including the thioredoxin–thioredoxin reductase system, thioredoxin peroxidases (peroxiredoxins), and glutaredoxins.^{208–210}

Other nonenzymatic antioxidants include glutathione, surfactant protein D, albumin, and proteins that bind iron and copper (powerful promoters of oxidative damage) such as transferrin, ferritin, ceruloplasmin, and lactoferrin.¹⁸⁶ Others include urate and bilirubin.¹⁹⁰

Diagnosis and Management of Pulmonary Oxygen Toxicity

Diagnosis

At present, there is no clinically useful means of diagnosing pulmonary O_2 toxicity in patients. Such a diagnostic capability would be most useful is ARDS. Physiologic indexes of pulmonary function (e.g., decrements in vital capacity), require patient cooperation and specialized equipment. But their greatest limitation is that in patients with ARDS such tests cannot distinguish between progressive O_2 toxicity and progression of the underlying cardiopulmonary condition. Chest radiographs also share this diagnostic limitation. Indexes of the lung's ability to metabolize biogenic amines (e.g., serotonin), polypeptides (e.g., by angiotensin-converting enzyme), and prostaglandins are sensitive in experimental animals but their use in critically ill patients is limited by their lack of specificity, dependence on capillary surface area and transit time, and the production by the injured lung of related compounds.²⁵ The pathology of pulmonary O_2 toxicity, diffuse alveolar damage, is a nonspecific (albeit clearly identifiable) pattern seen in ARDS from any cause. The sophisticated study by Montgomery et al,⁸⁰ which examined solute permeability of the lung, indexes of systemic toxicity, and pulmonary function, demonstrated that there is no better index of O_2 toxicity than the individual's subjective symptoms of retrosternal pain suggestive of acute tracheobronchitis. Studies such as theirs suggest that critical care physicians should maintain a heightened awareness and seek out such symptoms in patients on high levels of O_2 and include the possibility of pulmonary O_2 toxicity in the differential diagnosis of chest pain in such patients.

Exogenous Antioxidant Therapy for the Lung

The many approaches that have been taken in efforts to increase pulmonary antioxidant capacity have been reviewed.^{188–190,211} The major antioxidant strategies that have been evaluated for the lung include: (a) exogenous administration of normal intracellular enzymatic antioxidants, such as SOD, catalase, and GSH; (b) administration of synthetic antioxidant mimetics; (c) pharmacologic inhibition of free radical generating systems, such as by the iron chelator deferoxamine; (d) supplementation with nonenzyme antioxidants, such as vitamin E, *N*-acetylcysteine, and dietary supplementation with polyunsaturated fatty acid; and (e) agents to prevent the neutrophil-mediated amplification of the hyperoxic injury. All these approaches have been tried in vitro or in animal models, but only a few have been tried in human patients.

Adenovirus-mediated overexpression of MnSOD in mice protects against hyperoxia-induced acute lung injury and lethality²⁰³ and overexpressing ECSOD increases tolerance to hyperoxia.^{212,213} In adult baboons exposed to 96 hours of 100% O_2 , aerosolized recombinant human (rh) MnSOD improved oxygenation and alveolar epithelial histology, and decreased lung edema.^{214,215} Because in these latter reports the MnSOD was found extracellularly, Asikainen and White¹⁹⁰ speculated that the beneficial effect of rhMnSOD may have been scavenging ROS generated by activated inflammatory cells. In premature human infants, intratracheal rhCuZnSOD did not change the 28-day mortality or the incidence bronchopulmonary dysplasia.^{216,217} Among the surviving infants, however, those who had been treated with rhCuZnSOD had less frequent acute respiratory events (asthma medication requirements, emergency department visits, and hospitalizations) over the first year of life.²¹⁷

Synthetic SOD-mimetic (catalytic) antioxidants have been studied,^{188,190} particularly the metalloporphyrins. They are highly active, stable, potentially nontoxic, and protect against ROS-mediated lung injury in experimental models.¹⁹⁰ They scavenge superoxide, hydrogen peroxide, and peroxynitrite, and inhibit lipid peroxidation.²¹⁸ In a fetal baboon model of bronchopulmonary dysplasia, the catalytic antioxidant metalloporphyrin AEOL 10113, given by continuous intravenous infusion, improved histology and proinflammatory mediators, but not oxygenation.¹¹¹

A number of studies have been carried out in effort to enhance the GSH-dependent antioxidant capacity of the lung. *N*-acetylcysteine readily crosses into cells and, by undergoing deacetylation, forms cysteine, the rate-limiting amino acid in GSH biosynthesis. *N*-acetylcysteine acts both by directly reducing $O_2^{\cdot-}$, H_2O_2 , $OH^{\cdot}$, and $HOCl$ (hypochlorous acid), and by supporting GSH synthesis.²¹⁹ Its clinical use is well established in the treatment of acetaminophen-induced hepatotoxicity.²²⁰ In animal studies,^{221–226} *N*-acetylcysteine protects against oxidant lung injury. In human studies, however, disappointingly, a half dozen clinical trials of *N*-acetylcysteine in critically ill patients with acute lung injury, ARDS, or multiple organ failure have failed to show benefit in survival or oxygenation.^{227–231} In 1998, a clinical trial with procysteine in patients with acute lung injury or ARDS was stopped for lack of efficacy.²³² In very-low-birth-weight infants, supplementation of selenium (required for glutathione peroxidase) also failed to improve outcome.²³³

Lisofylline, an antiinflammatory compound that modulates the metabolism of lipids, including the proinflammatory linoleic acid, showed promise in preclinical studies but failed to show beneficial effects in a randomized controlled trial of 235 patients with acute lung injury or ARDS.²³⁴

Clinical trials of vitamin E supplementation failed to prevent bronchopulmonary dysplasia in premature newborns²³⁵ or hyperoxic lung injury in neonates.²³⁶ In contrast, vitamin A supplementation slightly decreased the incidence of bronchopulmonary dysplasia in extremely-low-birth-weight infants.²³⁷ In adult critically ill surgical patients, the combination of **vitamins** E and C reduced the incidence of organ failure and length of stay in the intensive care unit.²³⁸

Newborn rats have increased survival during hyperoxic challenge after dietary supplementation with polyunsaturated fatty acid (PUFA), including n-6 PUFA (linoleic and arachidonic acid, from safflower oil),²³⁹ n-3 PUFA (eicosapentaenoic and docosahexaenoic acid; from fish oil),²⁴⁰ and enteral Intralipid (commercial combination n-6 and n-3 PUFA; from soybean oil).²⁴¹ In ARDS, the combination of **vitamins** E and C plus fish oil (eicosapentaenoic acid) and borage oil (γ -linolenic acid) improved pulmonary neutrophil recruitment, gas exchange, requirement for mechanical ventilation, length of intensive care unit stay, and number of new organ failures.²⁴² A retrospective analysis of a subset of only forty-three of these patients²⁴³ reported that the twenty-one treated patients had decreased levels of interleukin-8, leukotriene B_4 , and neutrophils in bronchoalveolar lavage fluid, and decreased alveolar-capillary membrane permeability. A more recent randomized controlled trial in 100 patients with acute lung injury, reported that, compared to standard diet alone, adding eicosapentaenoic acid plus γ -linolenic acid and antioxidants improved oxygenation, compliance, and time on mechanical ventilation, but not mortality.²⁴⁴ In a randomized controlled trial of 165 mechanically ventilated patients with severe sepsis, this same dietary intervention improved oxygenation, time on mechanical ventilation, occurrence of new organ dysfunctions, time in the intensive care unit, and mortality.²⁴⁵ More recently, the ARDS Network OMEGA trial of enteral omega-3 (N-3) fatty acids, γ -linolenic acid, and antioxidants was stopped for futility after enrolling 272 acute lung injury patients, with a 60-day adjusted mortality of 24.6% versus 17.9% in control patients.²⁴⁶

In premature baboons the iron chelator deferoxamine to prevent hyperoxic lung injury (by inhibiting iron-catalyzed ROS generation) resulted in rapid cardiovascular collapse and death (<42 hours) in all animals.²⁴⁷

Another approach at lung protection from hyperoxic injury is to block neutrophil accumulation in the lung and thereby block further amplification and propagation of ROS-mediated injury.¹⁸⁸ Such blockade can be achieved by antibodies directed against cytokine-induced neutrophil chemoattractant-1 (CINC-1), which is the major rat neutrophil chemoattractant cytokine, or chemokine.²⁴⁸ This antibody, given intraperitoneally in a newborn-rat model of bronchopulmonary dysplasia, prevented neutrophil influx and preserved

lung compliance, alveolar development, and lung cell proliferation.²⁴⁸ Neutrophil influx can also be prevented by blocking the receptor, CXCR2, through which the chemokines act. In the same rat model, Auten et al²⁴⁹ showed that a competitive antagonist (SB-265610, a nonpeptide) of this chemokine receptor also prevented hyperoxia-induced neutrophil accumulation. Recent studies also show that other chemokines, including interleukin-8 and monocyte chemoattractant protein-1, play important roles in neutrophil attraction to the lung^{188,250} and are potential therapeutic targets. Mercer and Crapo¹⁸⁸ pointed out that the neutrophil-blocking approach and the ECSOD-enhancement approach appear to be complementary. That is, ECSOD enhancement protects against direct oxidant injury and neutrophil blockade prevents injury in areas not well protected by the enhanced ECSOD.

Several other experimental approaches have been investigated and reviewed in detail by Asikainen and White,¹⁹⁰ including catalytic antibodies, lazaroids, and novel approaches involving transcription factors, signaling pathways, DNA repair mechanisms, and growth factors. Despite their promise, none has yet been shown to be effective in humans.

Induction of Tolerance to Hyperoxia

In 1978, Frank et al²⁵¹ made the fortuitous discovery that endotoxin fully protected rats from hyperoxic morbidity and mortality. Similar results had been reported earlier following other toxic insults to the lung, including oleic acid,²⁵² α -naphthylthiourea,²⁵³ and diphosgene,⁷⁸ all of which were reported to increase survival on subsequent exposure to hyperoxia. Frank et al²⁵¹ also showed that endotoxin induced increases in SOD, catalase, and glutathione peroxidase. They speculated that in patients with acute respiratory failure (“for example... hyaline membrane disease,” or infant respiratory distress syndrome), coincident gram-negative infection may modulate the toxic effects of therapeutic hyperoxia.²⁵¹

In animals, other challenges that are protective against the lethality of subsequent exposure to hyperoxia include sublethal hyperoxia,^{254,255} hypoxia (in the absence of hypercarbia),^{256,257} and the combination of tumor necrosis factor- α and interleukin-1.²⁵⁸ Tumor necrosis factor- α and interleukin-1 are proinflammatory cytokines that mediate much of the pathophysiology of endotoxin.^{197,259,260} In rats the protective effects of each of these pretreatments can be explained by the induction of SOD, catalase, and glutathione peroxidase.^{251,254,257,261} The protective effect, however, may also occur in the absence of antioxidant enzyme induction. In the study by White et al,²⁶¹ the induction of the antioxidant enzymes induced by pretreatment with tumor necrosis factor- α plus interleukin-1 occurred relatively late (after 72 hours), after all control rats had already died. These authors suggested that the mechanisms of the lung protection may include, in addition to enzyme induction, the recruitment to the lung of more resistant cell lines.²⁶¹ Baker et al²⁵⁵ showed that the protective effect of preexposure to sublethal hyperoxia in adult rabbits, unlike that in neonatal rabbits²⁶² and rats, is not accompanied by an increase in antioxidant enzymes; it may be explained, instead, by the observed increase in endogenous surfactant induced by the preexposure to hyperoxia.²⁵⁵ More recently, additional mechanisms to explain the ameliorating effects on lung injury of cytokines or preexposure to hyperoxia have been provided by He et al²⁰⁰ (see Fig. 45-14 and “Cell Death by Hyperoxia: Apoptosis and Necrosis” above).

The clinical relevance of these observations is obvious. Critically ill patients with acute respiratory failure are commonly exposed to such “pretreatments,” namely, hyperoxia, hypoxia, endotoxin, and (endogenous) tumor necrosis factor- α and interleukin-1. Additionally, as part of routine nutritional support, these patients commonly receive nutritional supplementation with polyunsaturated fatty acid, as discussed above.

Exogenous Surfactant Administration

Surfactant is discussed in detail in Chapter 60 and is reviewed elsewhere.^{263,264} Endogenous surfactant is a complex mixture of approximately 85% to 90% phospholipids, 7% to 10% apoproteins (surfactant proteins A, B, C, and D), and 4% to 7% neutral lipids.²⁶³ Its primary function is to reduce the otherwise high liquid-gas surface tension within the alveoli and thereby decrease their tendency to collapse. Hyperoxia impairs surfactant function.²⁶⁵ Both ROS and RNS degrade surfactant, particularly the surfactant proteins, but the damage from RNS, especially peroxynitrite, appears more important.²⁶³ In most species, despite its impairment of surfactant function,

hyperoxia also induces surfactant synthesis.^{255,266,267} Endogenous surfactant has intrinsic antioxidant activity,^{268,269} can enhance intracellular antioxidant enzyme content,²⁶⁸ and in animal models, protects from hyperoxic lung injury and death.^{64,268,270}

Exogenous surfactants vary significantly in their composition.²⁶⁴ They also have antioxidant activity²⁷¹ and in animal models, protect against hyperoxic lung injury and death.^{271–274} After administration of exogenous surfactant (Curosurf), human preterm infants with respiratory distress syndrome show enhanced SOD and catalase activity in the tracheobronchial secretions.²⁷⁵

Exogenous surfactant therapy is now standard in neonatal intensive care units.^{264,276} A number of clinical trials in young patients have reported beneficial outcomes in newborns with respiratory distress syndrome,^{276–280} in infants with acute respiratory failure caused by respiratory syncytial virus,²⁸¹ and in children with hypoxemic acute respiratory failure.²⁸² In contrast, in adult patients with ARDS, the results have been disappointing.²⁶⁴ A trial of 725 adult patients with sepsis-induced ARDS using surfactant (Exosurf) found no benefit.^{283,284} The failure of this trial may be related in part to differences in both the surfactant constitutions and the delivery systems, between the neonatal-pediatric trials and the adult trials.²⁶⁴ Very recently, a trial of recombinant surfactant protein C-based surfactant in 843 adult patients with severe direct lung injury again showed no clinical benefit, perhaps related to insufficient surface activity.^{285,286}

Minimization of Oxygen Toxicity in the Intensive Care Unit

Ironically, in several ways the disease process of ARDS itself would seem to provide some degree of protection against hyperoxic injury. First, the alveolar epithelial and endothelial cells in the areas of shunt (presumably the areas of greatest injury) have, by definition, relatively low values of P_{O_2} , similar to the P_{O_2} of the mixed venous blood, and thus are not directly exposed to hyperoxia. Second, in ARDS the systemic circulation is also protected from hyperoxia because, by definition, this syndrome is characterized by systemic arterial hypoxemia. Third, the pathophysiology of ARDS and its management include exposure to hyperoxia, hypoxia, endogenous cytokines such as tumor necrosis factor- α and interleukin-1,^{287,288} and, commonly, endotoxin.^{287,288} However, as discussed above, prior exposure to each of these agents induces in animals tolerance and protection against future exposure to hyperoxia. Thus, patients who survive the initial insult of ARDS may be better able to withstand ongoing hyperoxic exposure. Fourth, the intraalveolar hemorrhage and exudation of plasma that is characteristic of ARDS (see discussion of pathology above) provides erythrocytes rich in antioxidant enzyme capacity, as noted above, and plasma proteins with substantial antioxidant activities. In fact, bronchoalveolar lavage fluid from ARDS patients possesses greater antioxidant activity than bronchoalveolar lavage fluid from normal subjects, largely as a result of the plasma proteins transferrin and ceruloplasmin.²⁸⁹

Once the possibility of pulmonary O_2 toxicity is considered (generally in an ARDS patient requiring $FI_{O_2} \geq 0.60$ for >24 to 48 hours), the only accepted treatment clinically at present is to reduce the FI_{O_2} to lowest level consistent with adequate systemic oxygenation. This goal should be approached by first minimizing excess O_2 demand (e.g., secondary to fever, infection, anxiety, pain, or “fighting the ventilator” with the judicious use of antipyretics,²⁹⁰ antimicrobials, sedatives, analgesics, or neuromuscular blocking agents,²⁹¹ respectively). Second, one should optimize all other determinants of systemic oxygenation, such as hemoglobin concentration, by using a restrictive strategy of red blood cell transfusion,^{292,293} and overall respiratory and cardiovascular status.²⁹⁴

As discussed in detail in other chapters, the best principles and practice of mechanical ventilation should be applied. These would include optimizing positive end-expiratory pressure, tidal volume, and plateau pressure using an open-lung or lung-protective ventilation strategy,² consideration of alternative modes of ventilation (e.g., high-frequency ventilation), prone ventilation,²⁹⁵ inhaled NO, extracorporeal membrane oxygenation, and so on.

Complicating conditions interfering with oxygenation should be meticulously sought out and treated. These conditions may include superimposed nosocomial pneumonia, excessive secretions, bronchospasm, occult pneumothorax and other parenchymal ventilator-induced lung injury, inadvertent intubation of a mainstem bronchus, occult large posterior pleural effusions, cardiopulmonary shunts unrelated to parenchymal lung injury (e.g., patent foramen ovale²⁹⁶), pulmonary emboli, cor pulmonale, low cardiac output, and

hydrostatic-pressure pulmonary edema (elevated pulmonary arterial wedge pressure). Hydrostatic-pressure pulmonary edema should be minimized by adopting a conservative strategy of fluid management²⁹⁷ and considering, in the hypoproteinemic patient, albumin plus furosemide therapy, which can improve fluid balance, oxygenation, and hemodynamics.²⁹⁸ Inotropic agents (e.g., dobutamine) should be considered as a possible means of decreasing the pulmonary arterial wedge pressure (and thus the hydrostatic gradient for pulmonary edema formation) while maintaining or increasing cardiac output. Supranormal cardiac indices, however, should be avoided.^{299,300}

After the above concerns have been adequately addressed, the FiO_2 should be adjusted to the lowest value required to maintain the arterial saturation within an acceptable range (e.g., 88% to 95%²). (Mountain climbers tolerate PaO_2 of 28 torr for brief periods³⁰¹ and SaO_2 of 73% to 85% for at least 4 days including with exercise.^{302,303}) Although the treatment of ARDS is largely supportive, treatment should, of course, be given for the underlying disease that may have produced the ARDS, particularly, if it is infection. In especially difficult patients with ARDS and refractory hypoxemia requiring sustained high levels of inspired O_2 , the clinician may be faced with the very difficult task of choosing among (a) accepting a lower level of arterial oxygenation, (b) increasing the already high airway pressures and risking ventilator-induced lung injury, and (c) continuing or even increasing the already high FiO_2 . In these very difficult cases it seems prudent to keep in mind that, based on information to date, the known immediate risks of systemic hypoxemia or ventilator-induced lung injury are clearly more established and devastating than the potential future risk of hyperoxia-induced lung injury. That is, the priorities should be, in order of their vital importance: (a) maintenance of arterial oxygenation; (b) avoidance of ventilator-induced lung injury; and (only then) (c) avoidance of high inspired O_2 fractions.

Important Unknowns and the Future

The most important unknown is the optimum inspired O_2 fraction to use in ARDS. Remarkably, the answer is still unknown more than four decades after ARDS was defined.³⁰⁴ Other unknowns concern the relationship between O_2 toxicity and ventilator-induced lung injury. These two entities have in common that both are a concern in acute respiratory failure, both share mechanisms of injury, and both initiate their injury in the lung but cause death from systemic multiorgan organ failure more than from their pulmonary injury. Future work should resolve their relative importance in acute lung injury: Are they independent or synergistic forces? Will therapy for one also be therapeutic for the other? In ARDS would there be further benefit in keeping the FiO_2 higher than usually done (e.g., near 100%) and thereby allowing use of an even more aggressive lung-protective strategy (that is, even lower tidal volumes and plateau pressures than those targeted in the ARDS Network trial)?² Future research should resolve how optimally to trade off the risks of hypoxemia, ventilator-induced lung injury, and O_2 toxicity. This trade-off will be facilitated when more specific markers of O_2 toxicity become available.

The degree to which patients with ARDS have induction of tolerance to hyperoxia (by preexisting hyperoxia, hypoxia, endotoxin, and cytokines; see above) is unknown. This induction could be manipulated to optimize a patient's antioxidant defenses. The potential benefit of combining two or more complementary therapies, for example, neutrophil blocking plus ECSOD enhancement,¹⁸⁸ are yet to be explored. Novel therapies, like activated protein C, which has been successful uniquely in severe sepsis,³⁰⁵ give hope that similar results can be achieved in ARDS and O_2 toxicity. Other proposed novel approaches involve transcription factors, signaling pathways, DNA repair mechanisms, and growth factors.¹⁹⁰ One of the most promising and recent approaches is to directly inhibit activation of the oxidant-mediated intrinsic apoptotic pathway of cell death (see Fig. 45-14).²⁰⁰⁻²⁰³

Summary and Conclusion

More than two centuries after his discovery of O_2 , Priestley seems remarkably prescient regarding oxygen's medical and toxic potential. O_2 is now recognized as lifesaving in patient conditions ranging from ambulatory to critically ill. The enormous expansion of knowledge about its biochemistry, physiology, and toxicology has relevance throughout all of biology. Continuing growth in our understanding of this "vital air" of Lavoisier is vital to continuing improvement in the care and safety of patients.

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